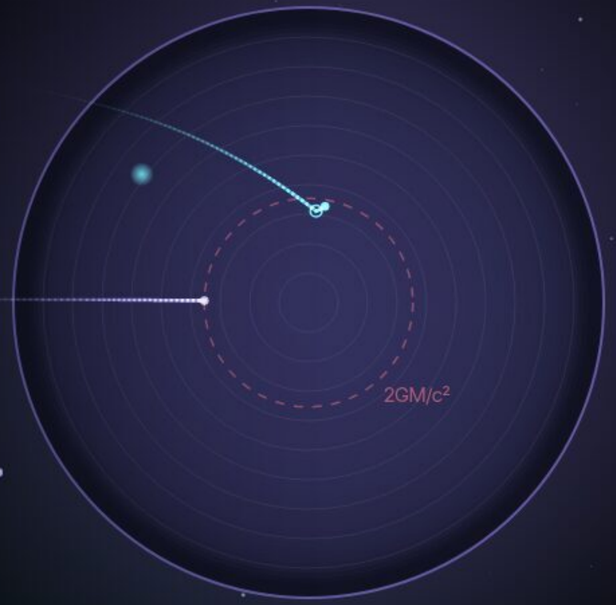
DARK MATTER IN NEUTRON STARS

Simulating the *invisible*



What we know about dark matter, how a neutron star captures it *one collision at a time*, and what state the captured particles end up in.

Rafael C. R. de Lima

Department of Physics · UDESC/CCT · Joinville, Brazil

with Jorge A. Rueda · ICRANet and University of Ferrara

$\mathscr{E} = \gamma\, e^{\nu(r)/2} < 1$ CAPTURED

rafael-lima.pages.dev/talks

OUTLINE

One problem, two papers, *one code*

I

The missing mass

How the problem was found, why the evidence holds, the ideas most explored today, and what the experiments have ruled out. Why look at a neutron star at all?

From Zwicky to the neutrino fog

II

OBSCURA: every collision

A relativistic Monte Carlo of capture that follows each particle through the interior metric, and reduces exactly to the optical-depth closure. What does the closure miss?

de Lima & Rueda, *A Monte Carlo relativistic framework for dark matter capture by neutron stars* (in preparation)

III

Degeneracy before self-gravity

After capture: a classical tracer, a degenerate Fermi gas, or a dark component? Two diagnostics, **S** for gravity and **Θ** for statistics, and a map of their thresholds.

de Lima & Rueda, *Degeneracy Before Self-Gravity: Dark Matter Regimes in Neutron Stars* (draft)

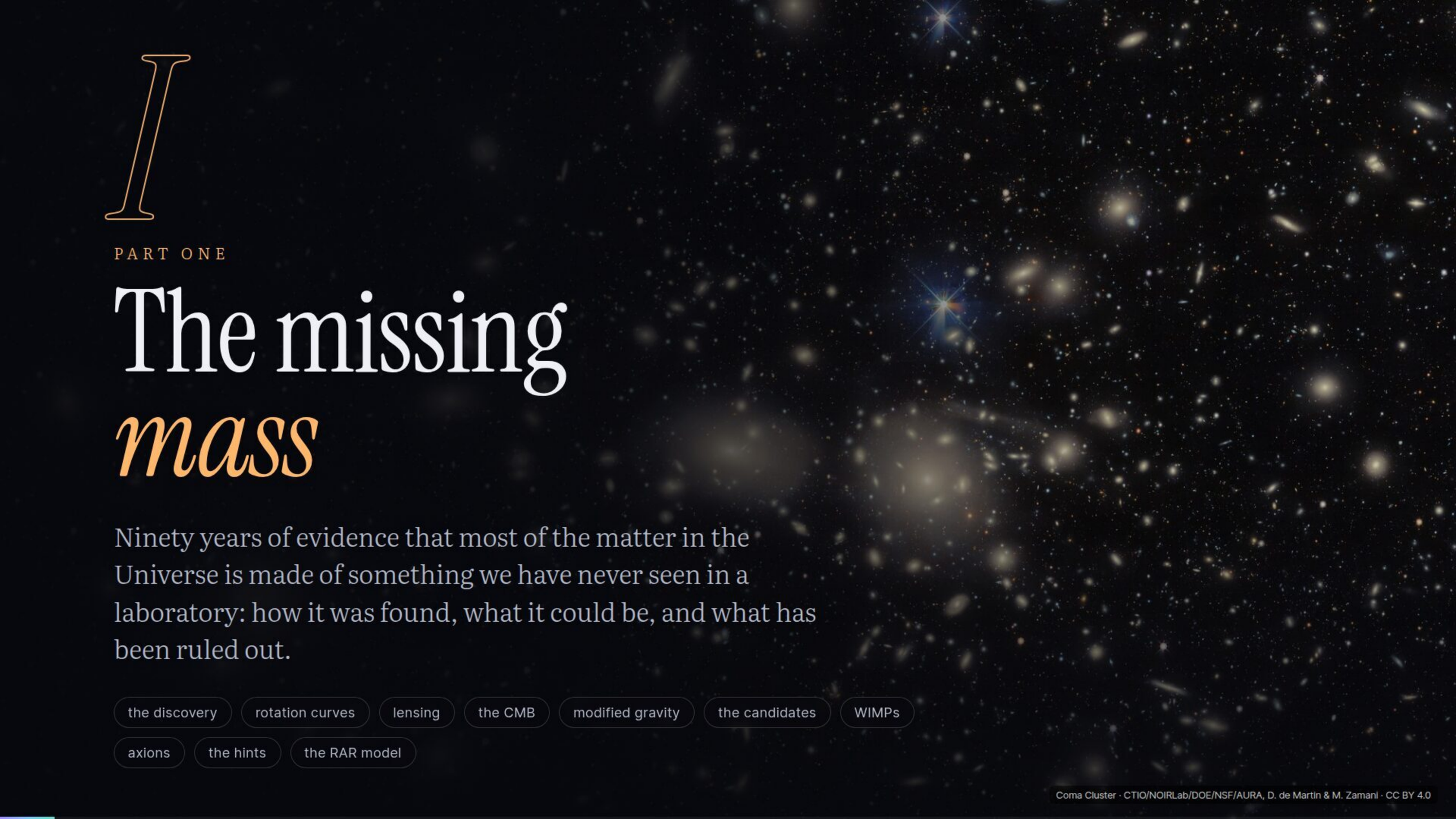
I

PART ONE

The missing *mass*

Ninety years of evidence that most of the matter in the Universe is made of something we have never seen in a laboratory: how it was found, what it could be, and what has been ruled out.

the discovery · rotation curves · lensing · the CMB · modified gravity · the candidates · WIMPs · axions · the hints · the RAR model



Coma Cluster · CTIO/NOIRLab/DOE/NSF/AURA, D. de Martin & M. Zamani · CC BY 4.0

1933 · THE COMA CLUSTER

Seven galaxies, *far too fast*



Fritz Zwicky

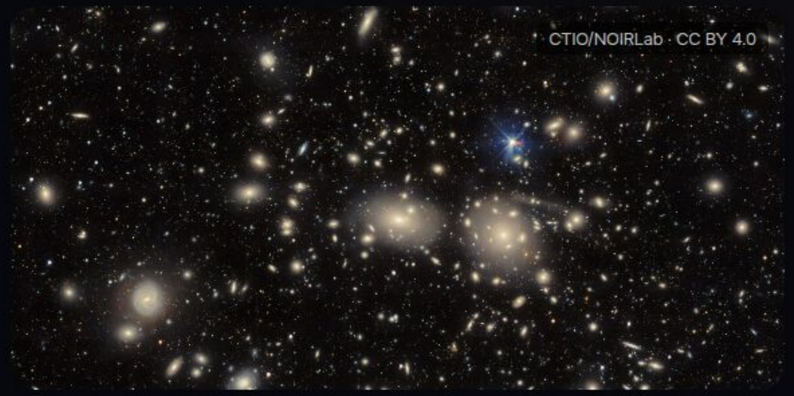
Helv. Phys. Acta 6, 110 (1933)

$$\langle v^2 \rangle = \frac{3}{5}\frac{GM}{R}$$

THE VIRIAL THEOREM, FOR A UNIFORM SPHERE

800 galaxies of 10^9 M$\odot$ within 10^{24} cm	80 km/s
Doppler spread he measured	≳ 1000 km/s
density required	≥ 400 × luminous

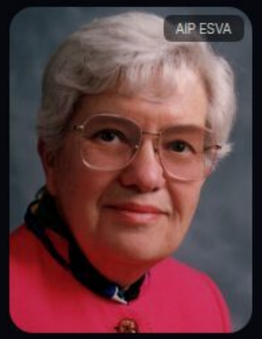
> *"...dark matter exists in much greater density than luminous matter."*
>
> Zwicky 1933, transl. H. Andernach (2017)



WHAT SURVIVED

Today's H_0 is seven times smaller than his, and X-rays later found hot gas outweighing the galaxies. The excess shrinks and stays: baryons are about **one sixth** of a cluster.

Galaxies that *refuse to slow down*





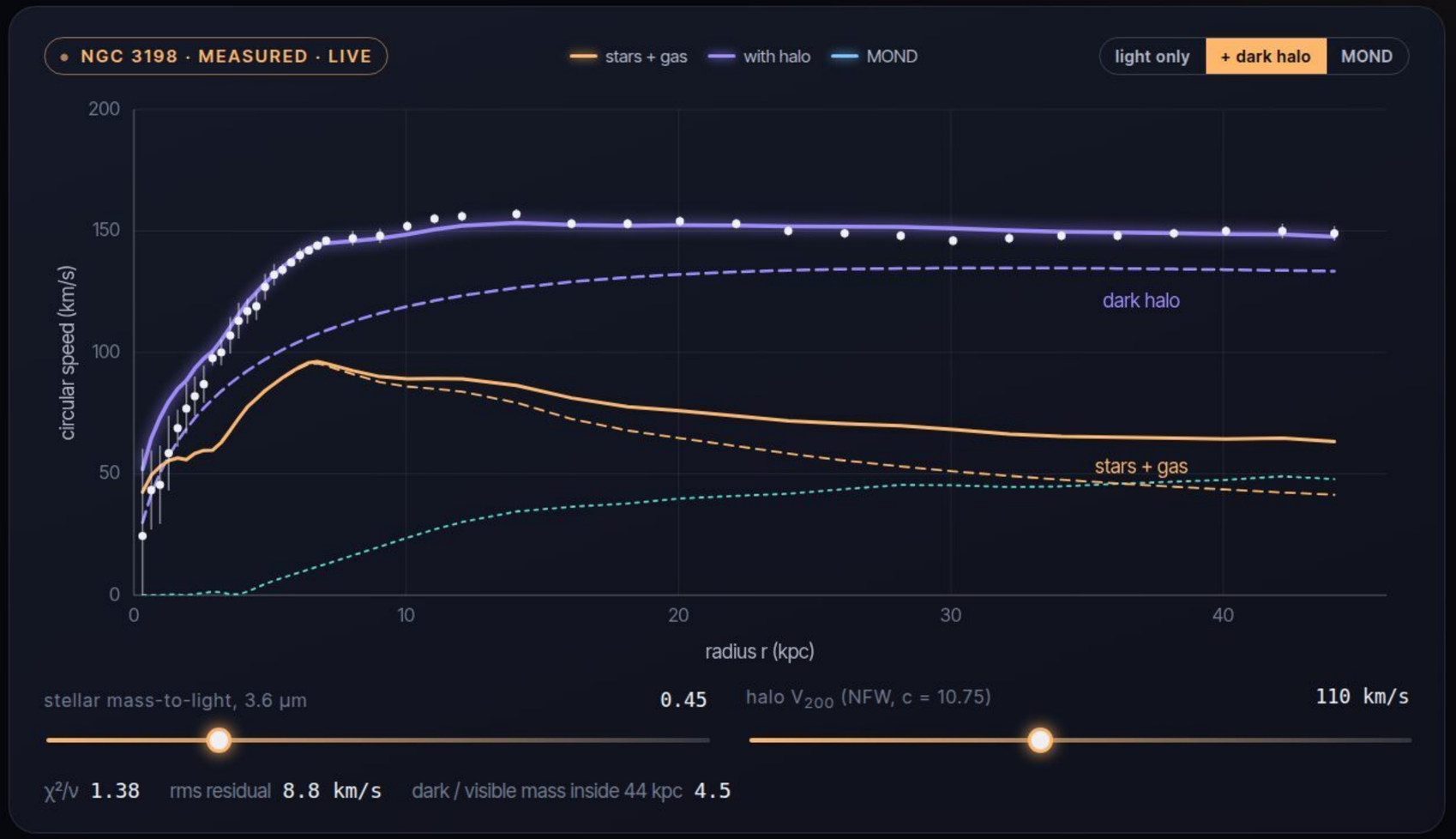
Vera Rubin

with Kent Ford, optical spectra from 1970; Albert Bosma in 21 cm, beyond the stars.

$$v^2(r) = \frac{GM(<r)}{r}$$

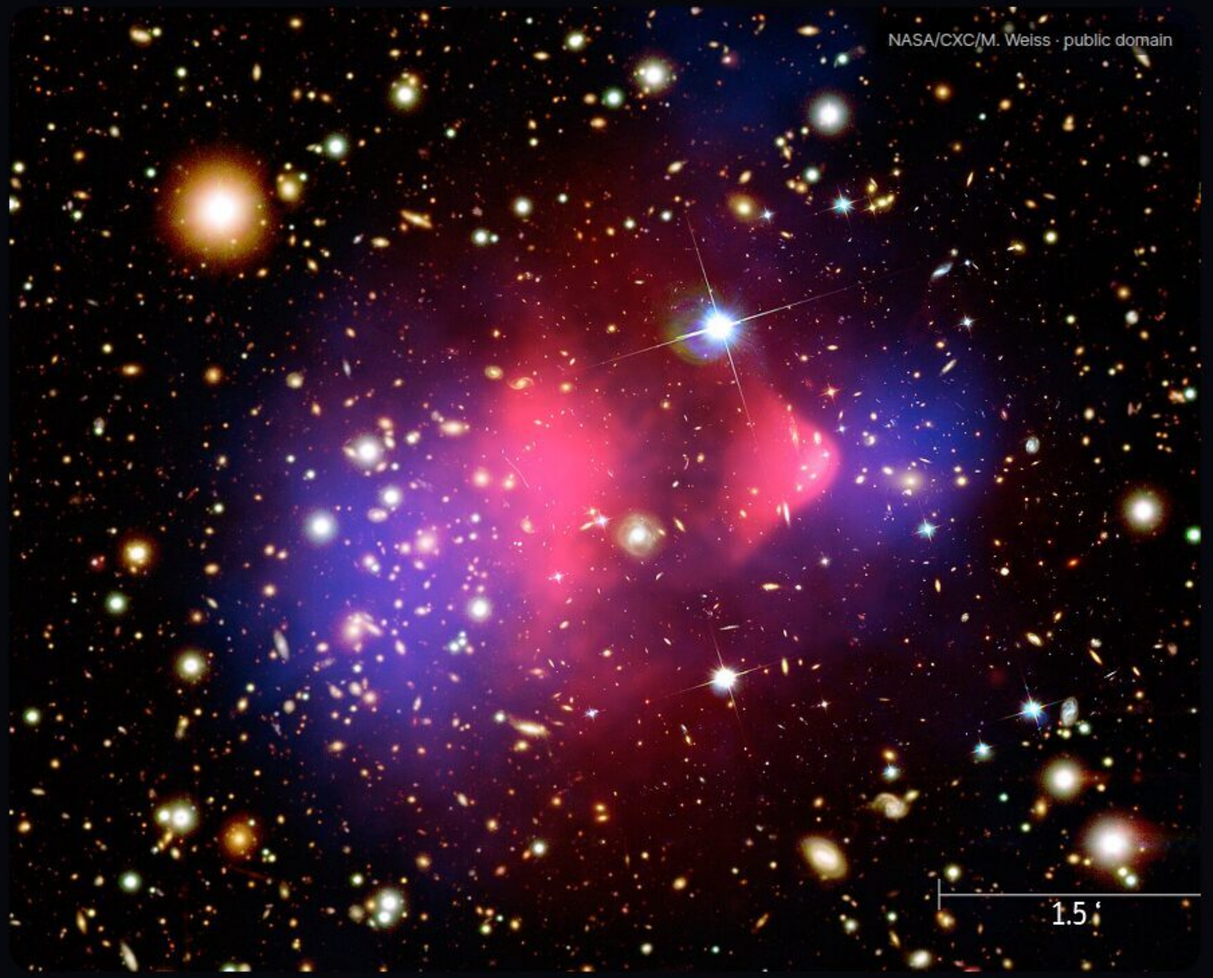
FLAT v MEANS M ∝ r

Rubin & Ford 1970; Rubin, Ford & Thonnard 1978, 1980; Bosma 1978. Data: SPARC, Lelli, McGaugh & Schombert, AJ 152, 157 (2016).



2006 · THE BULLET CLUSTER

Mass that *left the gas behind*



$$\theta_E^2 = \frac{4GM}{c^2}\,\frac{D_{ls}}{D_l D_s}$$

LENSING WEIGHS MASS, LUMINOUS OR NOT

Two clusters went through each other. The gas (pink, X-rays), which holds most of the baryons, was braked. The lensing mass (blue) kept going with the galaxies.

8σ OFFSET

Between the total mass and the baryons, and it “cannot be explained with an alteration of the gravitational force law.” Clowe et al. 2006

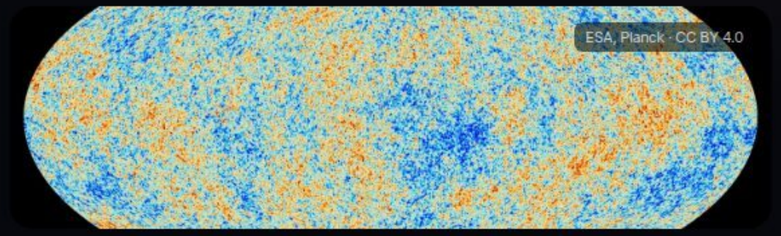
72 COLLISIONS LATER

Dark matter stays with the stars: $\sigma/m < 0.47\ \mathrm{cm^2/g}$ at 95%. It barely collides, even with itself. Harvey et al. 2015

380 000 YEARS AFTER THE BIG BANG

The early Universe *weighs it*





Baryons load the photon fluid: compression peaks rise over rarefaction peaks.

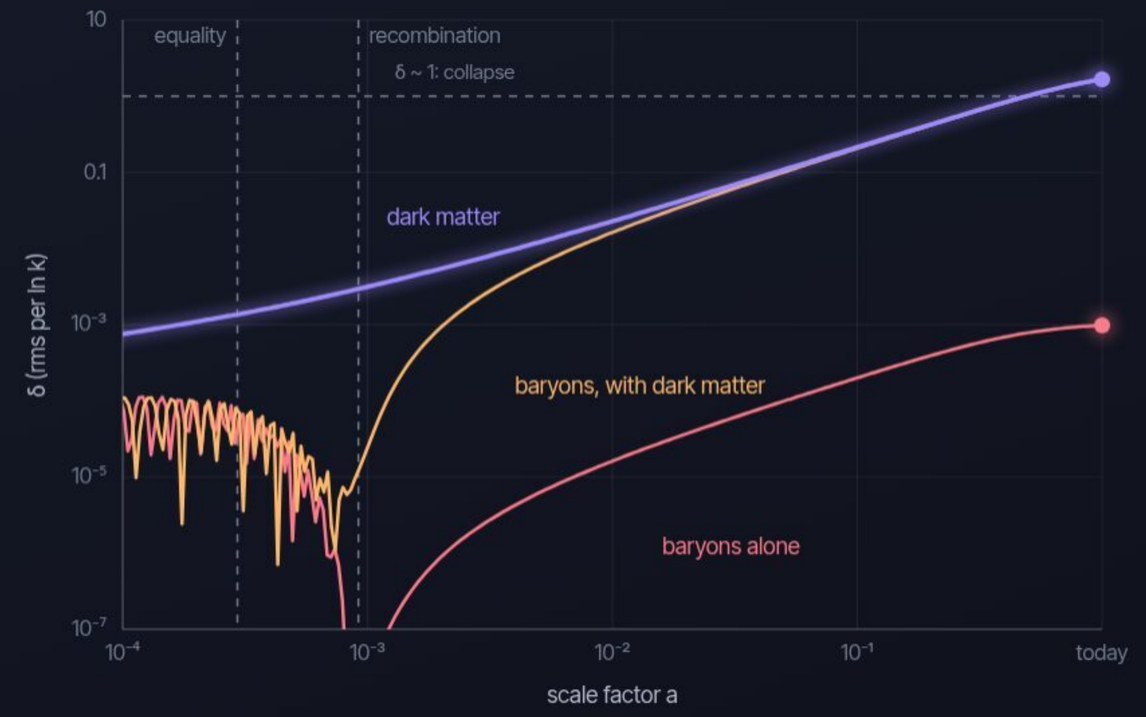
Dark matter feels no pressure, so its wells do not decay. Without it, the first peak more than doubles.

$$\Omega_c h^2 = 0.120 \pm 0.001$$
$$\Omega_b h^2 = 0.0224 \pm 0.0001$$

PLANCK 2018

THREE MINUTES · 380 000 YEARS · TODAY

It cannot be *atoms*



Same primordial fluctuations. With dark matter, $\delta \approx 1.7$ today; with baryons alone, $\delta \approx 10^{-3}$: no galaxies.

TWO CLOCKS AGREE ON THE BARYONS

Primordial deuterium, $(D/H)_p = (2.527 \pm 0.030) \times 10^{-5}$, fixes the baryon density at three minutes. It matches the CMB, 380 000 years later, within 2σ.



atoms 4.9%, so dark matter is about 85% of all matter · Planck 2018, h = 0.674

NOT NEUTRINOS EITHER

$\Sigma m_\nu < 0.12$ eV makes them at most about **1%** of the dark matter, and too hot. **Nothing in the Standard Model fits.**

THE ALTERNATIVE

Or is *gravity* wrong?

$$a = (a_N\, a_0)^{1/2} \text{ for } a_N \ll a_0$$
$$v^4 = GM\, a_0$$

MOND, MILGROM 1983: $a_0 \approx 1.2\times10^{-10}$ m/s²

The regularity is real: in rotating galaxies, the baryons alone predict the acceleration. The MOND button in the rotation-curve panel uses exactly that relation, with no halo.

Any dark-matter theory has to **explain** it. But a theory of gravity has to work on **every** scale.

SCALE	TEST	
Galaxies	2693 points in 153 galaxies follow one relation with one constant	WORKS
Wide binaries	Gaia DR3: Newton preferred at 19σ (Banik et al.), an anomaly at low *g* (Chae)	DISPUTED
Clusters	Still short of mass by about a factor of 2, and in the Bullet the potential follows the galaxies	FAILS
GW170817	Gravity and light arrive together: the end of "dark-matter emulators"	FAILS
CMB, *P*(*k*)	AeST fits both, with a new field that behaves as dust in the early Universe	AT A PRICE

We know how much, *not what*

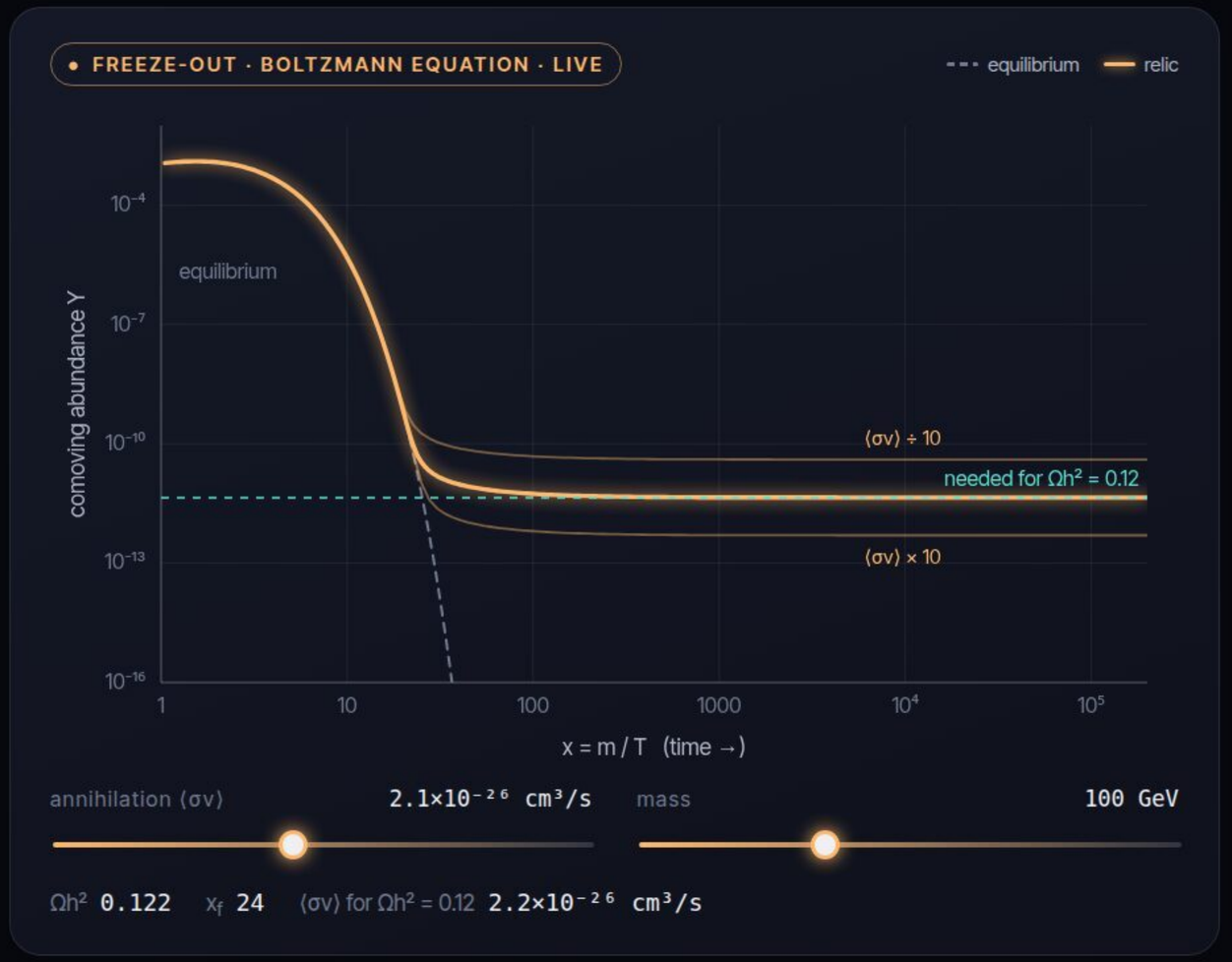
non-baryonic · cold · neutral · stable over cosmic time · nearly collisionless · $\Omega_c h^2 = 0.120$ · $\rho_\chi \approx 0.4$ GeV/cm^3 near the Sun



Bounds: ultralight scalars $m > 2\times10^{-20}$ eV (Rogers & Peiris 2021) · Pauli, any fermion $m \geq 0.13$ keV (Alvey et al. 2021) · thermal warm dark matter $m > 5.7$ keV (Iršič et al. 2024) · compact objects 10^{-8}–10^{3} M⊙ not the dominant component (OGLE, Mróz et al. 2024, 2025) · asteroid-mass window open (Gorton & Green 2024). Bright bands: the most explored today.

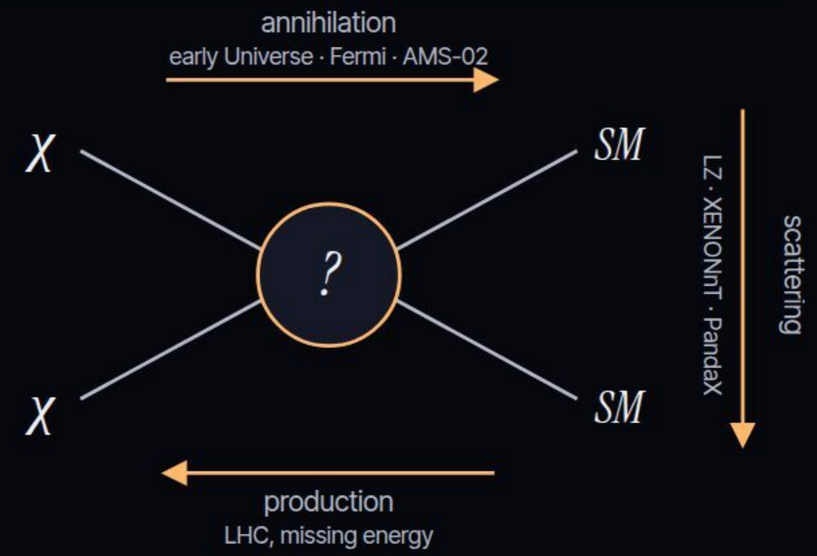
THE MOST EXPLORED IDEA

The *WIMP* miracle



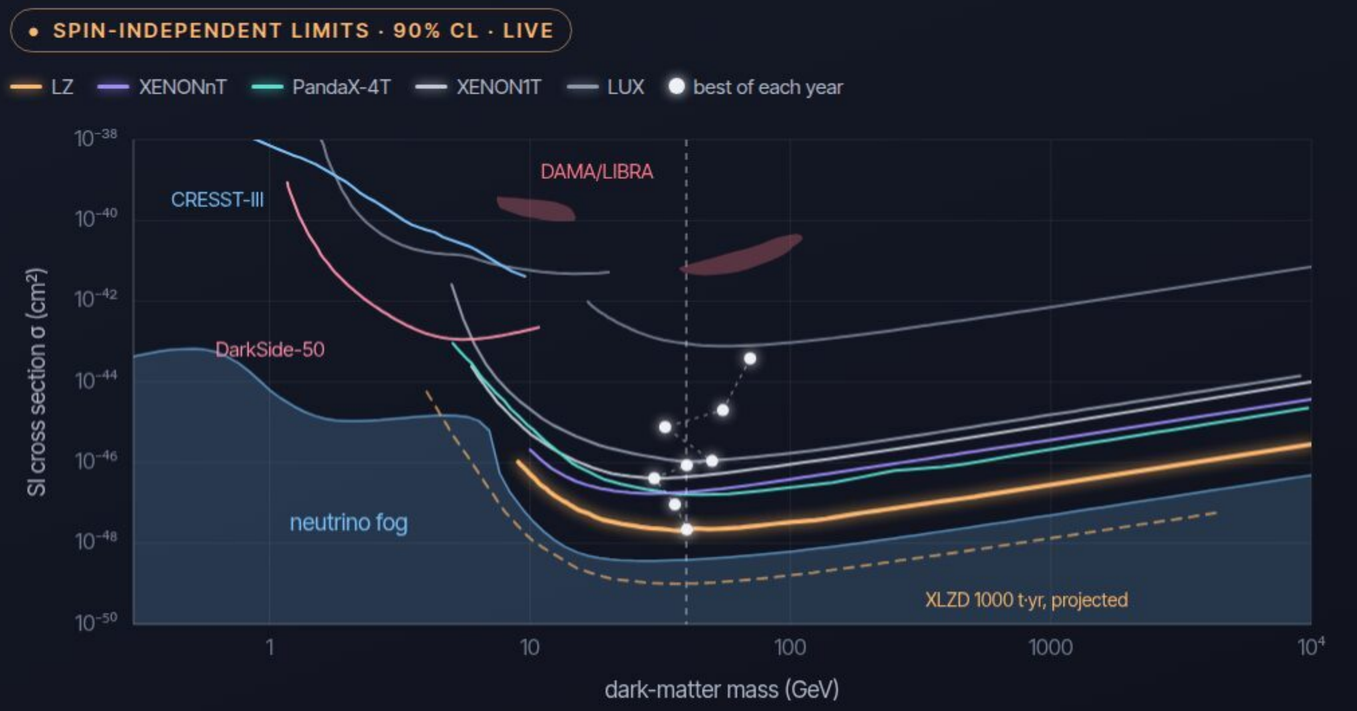
$$\frac{dY}{dx} = -\frac{\lambda}{x^2}(Y^2 - Y_{eq}^2)$$

THE RELIC DOES NOT DEPEND ON THE MASS, ALMOST



$\Omega h^2 = 0.12$ needs $\langle\sigma v\rangle \approx 2.2\times10^{-26}$ cm³/s above 10 GeV (Steigman, Dasgupta & Beacom 2012).

Four orders of magnitude, *no WIMP*



year 2026 mass 40 GeV

2026: best 2.2×10^{-48} cm² (LZ, 40 GeV) limit at this mass 2.1×10^{-48} cm² fog there 3.9×10^{-49} cm²
largest recoil on Xe 99 keV

$$\frac{dR}{dE_R} \propto \frac{\rho_\chi \sigma A^2 F^2(E_R)}{m_\chi \mu^2} \int \frac{f(v)}{v} d^3v$$

A²: HENCE TONNES OF XENON

THE FOG HAS ARRIVED

LZ, 5.7 t·yr: **^{8}B solar neutrinos** scattering coherently on xenon, at **4.5σ**. The next limits must separate them from a signal.

THE BLIND SPOT

Recoils fall as $\mu^2 v^2 / m_A$. Below a few GeV they drop under threshold: at 100 keV, about **1 μeV** on xenon.

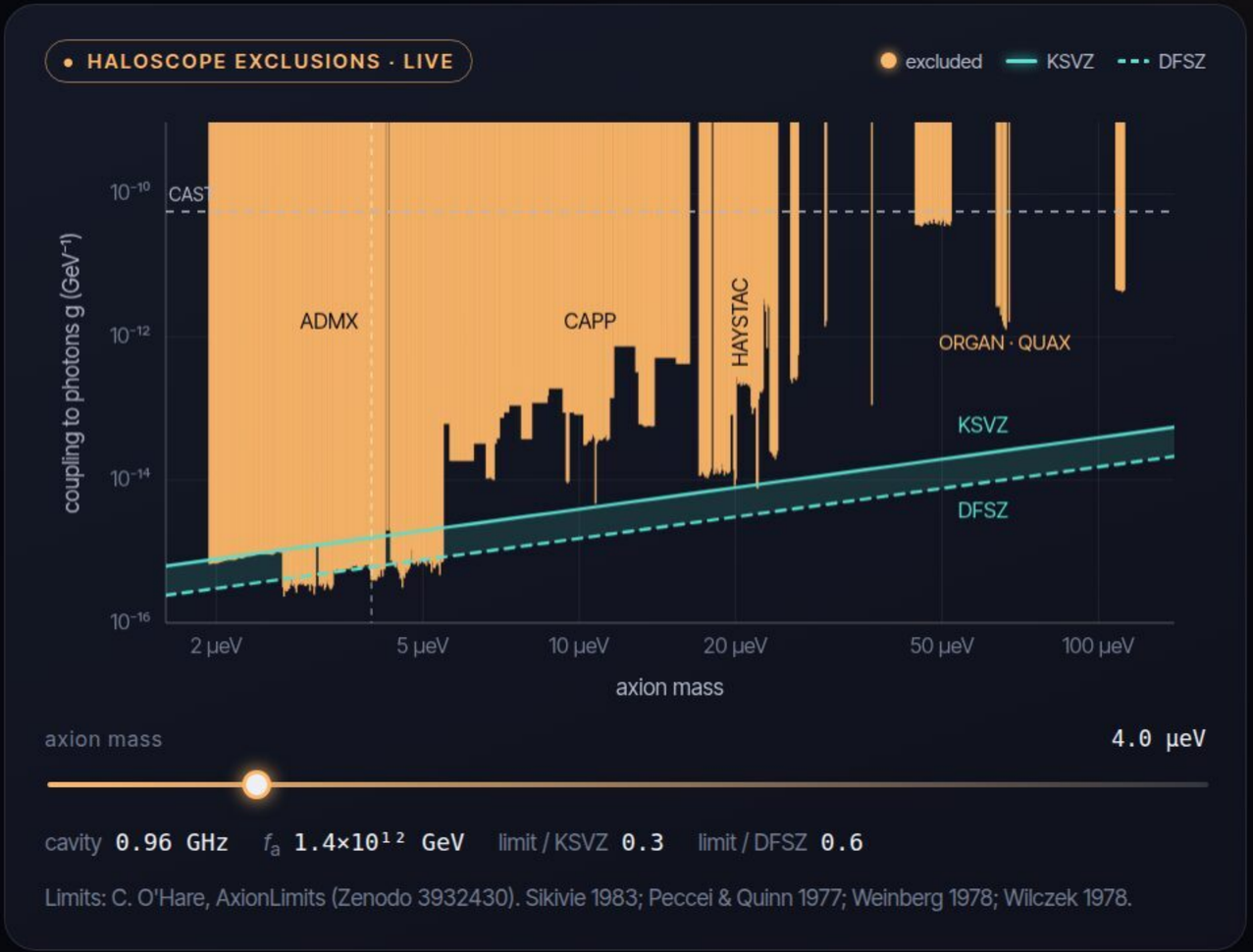
Limits: compiled by C. O'Hare. Fog: O'Hare, PRL 127, 251802 (2021). Best points: the papers' abstracts.

A particle invented *for another problem*

1. QCD allows a CP-violating angle $\bar{\theta}$, yet the neutron's dipole moment gives $|\bar{\theta}| \lesssim 10^{-10}$.
2. Peccei–Quinn: make $\bar{\theta}$ a field that relaxes to zero. Its quantum is the **axion**.
3. Oscillating coherently since the QCD epoch, it behaves as **cold dark matter**.
4. In a magnetic field $a \to \gamma$: a tuned cavity collects photons of $h\nu = m_a c^2$.

$$m_a \approx 5.7\ \mu\text{eV}\ \frac{10^{12}\ \text{GeV}}{f_a}$$

Haloscopes reach the benchmark **DFSZ** line only around **3–5 μeV**, a tenth of the range scanned.



Limits: C. O'Hare, AxionLimits (Zenodo 3932430). Sikivie 1983; Peccei & Quinn 1977; Weinberg 1978; Wilczek 1978.

Hints, and *what became of them*

CLAIM	SINCE	WHERE IT STANDS	
DAMA/LIBRA annual modulation	1998	Same NaI target, no modulation: COSINE-100 + ANAIS-112 exclude it at 4.7σ (1–6 keV) and 3.5σ (2–6 keV)	REFUTED
XENON1T electron excess	2020	A 3.4σ solar-axion fit; gone in XENONnT, with a background five times lower	GONE
3.5 keV X-ray line	2014	XRISM sees no line in ten clusters; its limit is still five times above the XMM detection	FADING
Galactic Centre γ-ray excess	2009	The photons are real; recent work leans to unresolved millisecond pulsars, but it is not settled	OPEN
20 GeV Milky Way halo excess	2025	Reproduced in Fermi data; in tension with dwarf galaxies and, in a preprint, with AMS-02 antiprotons	OPEN
LHC missing energy	Run 1–3	No established excess; no new particle at the weak scale so far	NOTHING

The cleanest indirect targets stay dark too: dwarf galaxies put $\langle\sigma v\rangle$ **below the thermal value** for $m \lesssim 100$ GeV in quark and τ channels (Fermi-LAT 2015). Status as of November 2026.

What if dark matter is *a Fermi gas?*

Take neutral fermions, gravity and the Pauli principle, and solve Einstein's equations at finite temperature. *Nothing else.*

1939 — **Oppenheimer & Volkoff**
Pauli against gravity: a limit mass

1969 — **Ruffini & Bonazzola**
self-gravitating quantum particles

1983–92 — **Ruffini & Stella; Ingrosso et al.**
semi-degenerate, energy cutoff

2002 — **Bilić et al.**
a Newtonian fermion ball at the Galactic Centre

2015 — **Ruffini, Argüelles & Rueda**
core, plateau and halo in GR

2018 — **Argüelles et al.**
the cutoff: a core at Sgr A*, a halo for the Galaxy

ONE PARTICLE

One distribution function and one fermion mass, from milliparsecs to a hundred kiloparsecs.

NO NEW FORCE

The pressure comes from quantum statistics and temperature. No interaction beyond gravity is needed.

THE MASS SHOWS

Unlike cold dark matter, the particle mass sets the structure: a degenerate core shrinks as $m^{-8/3}$.

Fermi–Dirac, *with a way out*

$$f_c(\varepsilon \le \varepsilon_c) = \frac{1 - e^{(\varepsilon - \varepsilon_c)/kT}}{e^{(\varepsilon - \mu)/kT} + 1}, \quad f_c(\varepsilon > \varepsilon_c) = 0$$

FERMI–DIRAC, EMPTIED ABOVE THE ESCAPE ENERGY

$$\rho = m \frac{2}{h^3} \int f_c \, (1 + \varepsilon/mc^2) \, d^3p, \quad P = \frac{1}{3} \frac{2}{h^3} \int f_c \, p \, v \, d^3p$$

THE EQUATION OF STATE, AT EVERY RADIUS; TOV CLOSES THE SYSTEM

Not a guess: a maximum-entropy state of collisionless relaxation with evaporation.

Lynden-Bell 1967; Chavanis 2004; Ruffini, Argüelles & Rueda 2015; Argüelles et al., PDU 21, 82 (2018).

TOLMAN

$e^{\nu/2}\,T$ = const: hotter deeper in the well

KLEIN

$e^{\nu/2}\,(\mu + mc^2)$ = const: more degenerate deeper in

ENERGY ALONG A GEODESIC

$e^{\nu/2}\,(\varepsilon_c + mc^2)$ = const: one conserved escape energy

$$\beta = \frac{kT}{mc^2}, \; \theta = \frac{\mu}{kT}, \; W = \frac{\varepsilon_c}{kT} \Rightarrow W(r) = W_0 + \theta(r) - \theta_0$$

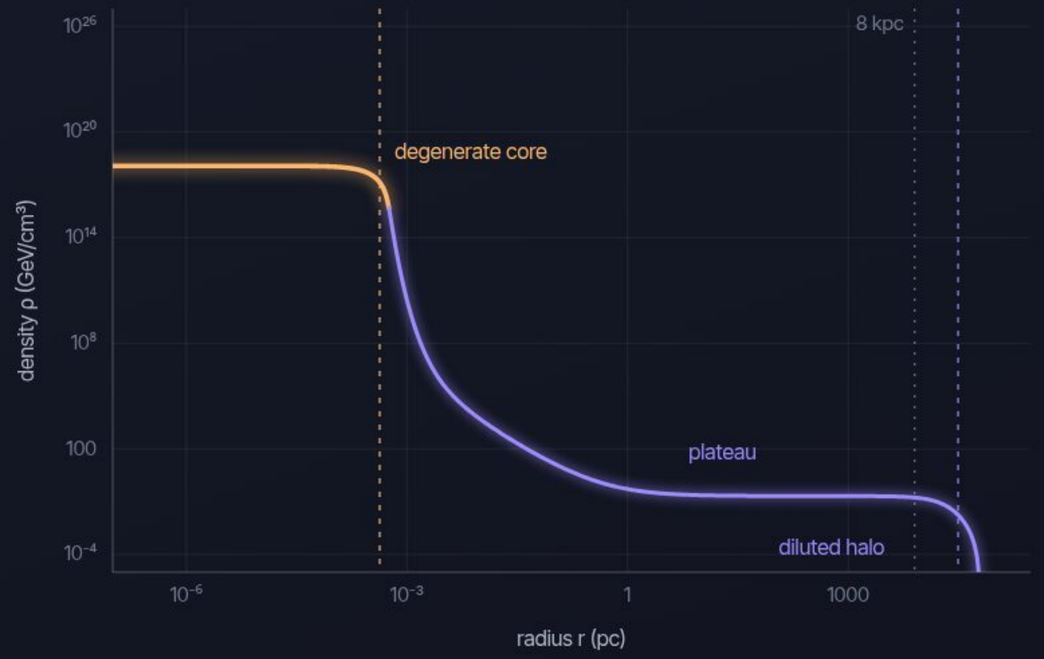
FREE: m, β_0, θ_0, W_0

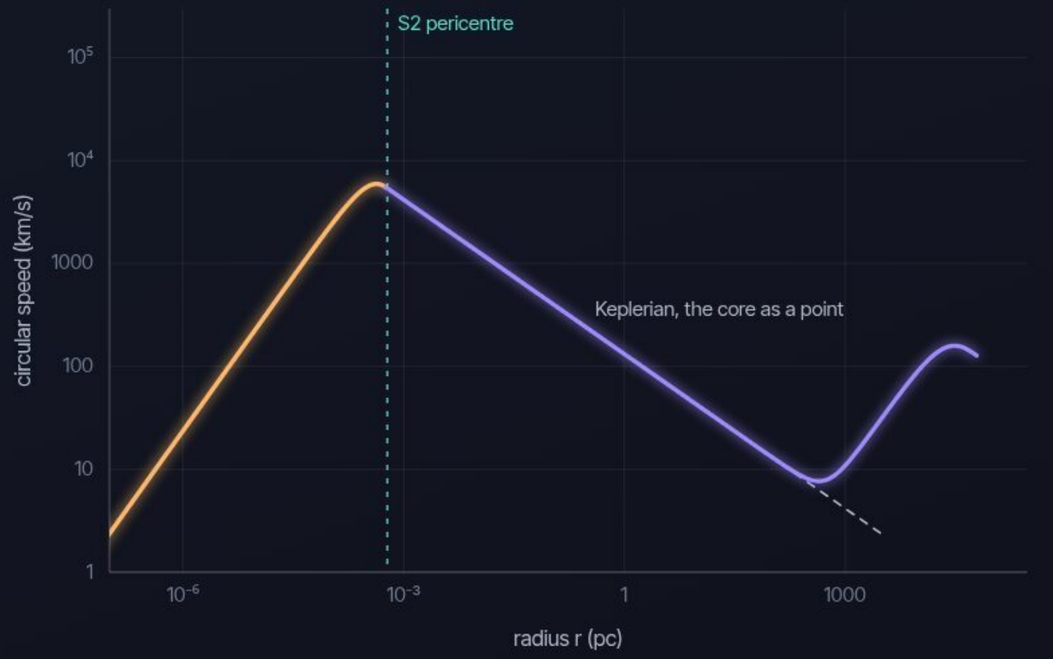
One gas, *three regimes*

TOV + FERMI–DIRAC · SOLVED IN YOUR BROWSER

FERMION, KEV: 48 | 56 | 100 | 345

RIGHT: rotation | θ and W | f(v) at 8 kpc





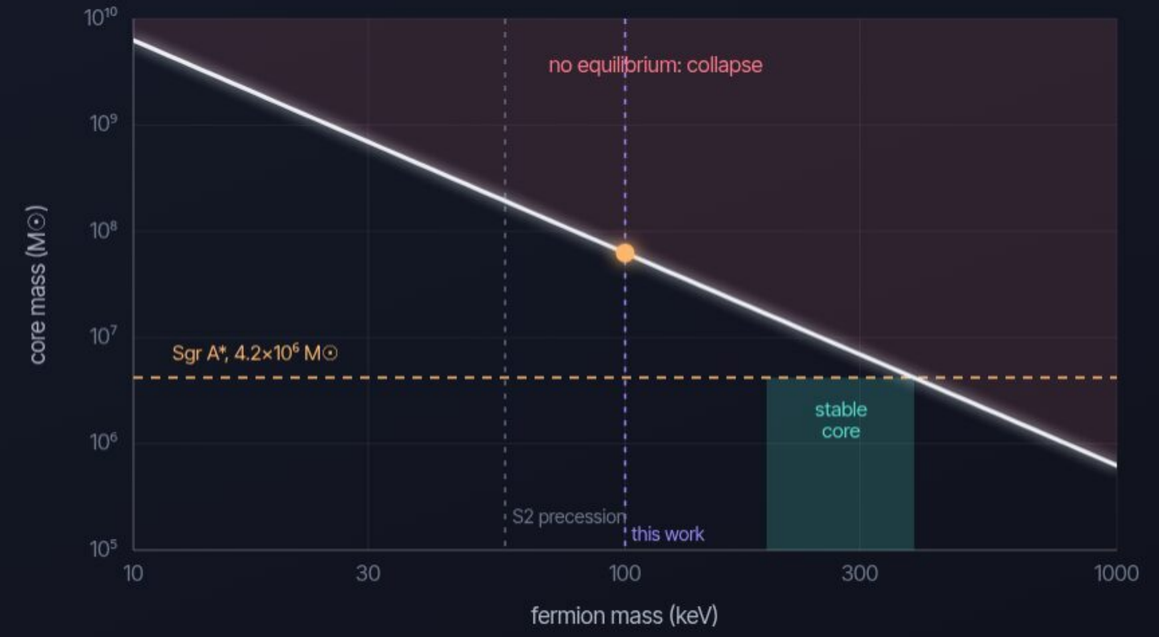
central degeneracy θ_0 37.77

central temperature β_0 1.20×10^{-5}

halo escape speed, $c\sqrt{(2W\beta)}$ 282 km/s

core 3.5×10^{6} $M_\odot$ in 0.427 mpc · seen from outside 4.05×10^{6} $M_\odot$ · halo peak 159 km/s at 31.2 kpc · edge 62.0 kpc · total 2.3×10^{11} $M_\odot$ · ρ(8 kpc) 0.17 GeV/cm³ · W_0 66.341

A neutron star, *rescaled*



fermion mass **100 keV**

limit **6.26×10⁷ M⊙** its radius **5.61 AU** central density **0.46 g/cm³**

a Sgr A* ball, non-relativistic **0.12 mpc** (S2 pericentre 0.6)

$$M_{\mathrm{OV}} = 0.384\,\frac{m_{\mathrm{P}}^{3}}{m^{2}}$$

ONE FORMULA, ONE NUMBER FOR EACH FERMION

IDEAL FERMIONS	NEUTRONS	100 KEV
Particle	939.6 MeV	100 keV
Limit mass	0.71 M⊙	6.3×10⁷ M⊙
Radius there	9.5 km	5.6 AU
Central density	3.6×10¹⁵ g/cm³	0.46 g/cm³
Radius / R_S	4.5	4.5

Lighter than water, and at the edge of collapse. Past the limit, a **black-hole seed** with no star involved.

Argüelles et al. 2018; Krut, Argüelles & Chavanis 2025; Argüelles et al. 2024.

A black hole, *or a core?*

TEST	RESULT	
Rotation curve	One continuous profile, with baryons, from the core to the Gaia DR3 curve	WORKS
S2 and G-source orbits	Core and black hole differ by less than 1% in orbital parameters; no conclusive preference	UNDECIDED
S2 precession	Retrograde below 56 keV, prograde above. GRAVITY's prograde signal points to $m \gtrsim 57$ keV, with under 0.1% of the core inside the orbit	BOUNDS m
Stability	Stable core–halo solutions with a 4.2×10^6 M☉ core exist only for **194–387 keV**	BOUNDS m
Images	Central depression and a ring, down to about 35 μas, but **no photon ring**: the key difference from a black hole	NEXT TEST
Collapse	Cores past the limit become black-hole seeds, sooner if baryons accrete onto them	PREDICTION

Our **100 keV** is a capture benchmark, not a claim about Sgr A*: it sits inside the 48–345 keV of the halo fits, below the stable window for a core that replaces the black hole. Status as of November 2026.

Argüelles et al. 2018; Becerra-Vergara et al. 2020, 2021; Argüelles et al. 2022; Pelle et al. 2024; Krut, Argüelles & Chavanis 2025; Crespi et al. 2026.

THE PREMISE

A neutron star is a *dark-matter detector*

As it crosses the Galaxy, a neutron star sweeps up dark matter: its potential well *focuses* the particles, its matter *stops* them.

halo → focusing → collisions → **capture** → accumulation → regime → heating · collapse · structure

A DEEP WELL

Escape speed at the surface $\sqrt{(2GM/R)}$ = **0.59 c** for 1.4 $M_\odot$ and 11.7 km. Slow halo particles are bent toward the star from a thousand radii away.

A DENSE TARGET

$n_b \approx 5\times10^{38}$ cm^{-3} at the centre: with $\sigma = 10^{-44}$ cm^2 a particle meets $\tau \approx$ **6–9** collisions in one crossing.

AN OLD, COLD STAR

What accumulates can heat it, collapse into a black hole, or change its structure — **if** we know how much is captured and in what state.

II

PART TWO

OBSCURA: *every collision*

A relativistic Monte Carlo that follows each dark-matter particle through the interior metric of the star, collision by collision — and recovers the optical-depth closure when its extra physics is switched off.

the halo · focusing · the optical-depth closure · the transport · what the closure misses · count laws · cooling



Bullet Cluster, 1E 0657–56 · NASA/CXC/M. Weiss · public domain

Two ways to fill *the halo*

STANDARD HALO · MAXWELL–BOLTZMANN

$$f(v) \propto v^2 e^{-v^2/v_0^2},\ v < v_{\mathrm{esc}}$$

v_0 = 220 km/s, v_{esc} = 544 km/s

The usual WIMP benchmark: particles of GeV to TeV, a few hundred km/s, isotropic. Used for the Maxwell–Boltzmann results of Part III.

RAR FERMIONS · TRUNCATED FERMI–DIRAC

$$f \propto \frac{1 - e^{(E - E_c)/kT}}{e^{(E-\mu)/kT} + 1},\ E \le E_c$$

RUFFINI–ARGÜELLES–RUEDA

Self-gravitating fermions in general-relativistic equilibrium: a degenerate core at the Galactic centre and a Boltzmann halo. Rotation curves and the S-stars allow **48–345 keV**; we take **100 keV**.



FORTY YEARS OF CAPTURE

From the Sun to *neutron stars*

1985
Press & Spergel
capture of WIMPs by the Sun

1987
Gould
the capture-rate formalism

1989
Goldman & Nussinov
WIMPs and neutron stars

2008–10
Kouvaris; Kouvaris & Tinyakov
heating by annihilation; collapse to a black hole

2017
Baryakhtar et al.
dark kinetic heating

2020
Bell et al.
relativistic capture with Pauli blocking, opacity, multiple scattering

2026
OBSCURA
collision histories through the interior metric

The orbit must *reach the surface*

1. Outside the star, two constants: $\mathscr{E} = (1 - 2M/r)\,u^t$, $\mathscr{L} = r^2 u^\varphi$. From infinity, $\mathscr{E} = \gamma$ and $\mathscr{L} = \gamma v b$.
2. $u_\mu u^\mu = -1$ leaves one dimension: $(u^r)^2 = \mathscr{E}^2 - V^2$, with $V^2 = (1 - 2M/r)(1 + \mathscr{L}^2/r^2)$.
3. No turning point before the surface: $\mathscr{E}^2 \geq V^2(R)$, with $C = M/R$.
4. That caps the angular momentum: $\mathscr{L}_c = R\,[(\mathscr{E}^2 - 1 + 2C)/(1 - 2C)]^{1/2}$.

$$\frac{b_{max}^{\,2}}{R^2} = \frac{1}{1-2C}\left[1 + \frac{2C}{(\gamma v)^2}\right] \text{ vs Newton } 1 + \frac{2C}{v^2}$$

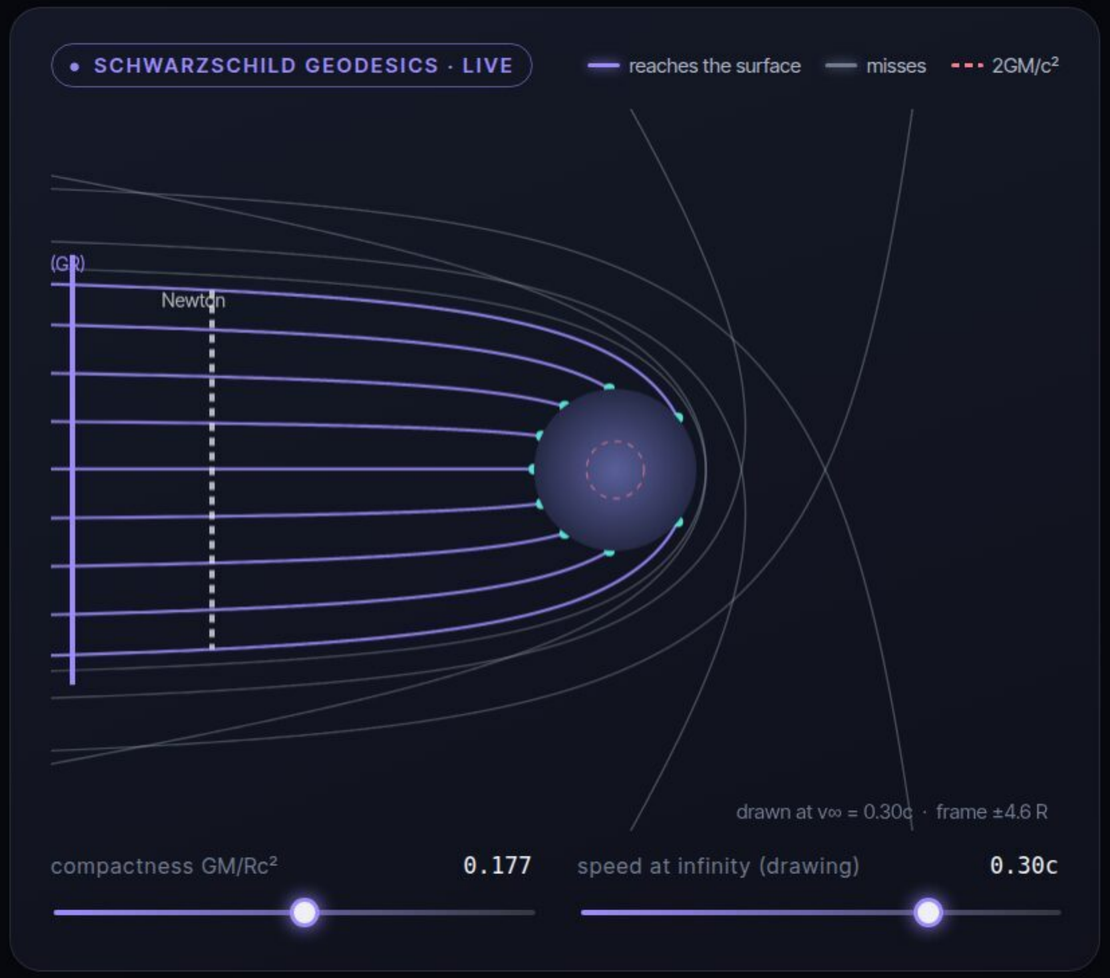
TWO CORRECTIONS: v → γv, AND THE APPARENT RADIUS R/√(1 − 2C)

Exact for $C \leq 1/4$. Above that, slow particles meet the barrier **outside** the star and $\mathscr{L}_c \to 4M$.



GRAVITATIONAL FOCUSING

The star looks *a thousand times wider*





$$\frac{b_{\max}}{R} = \frac{1}{\gamma v}\sqrt{(\gamma^2 - 1 + 2C)\,/\,(1 - 2C)}$$

FROM $\mathscr{L} < \mathscr{L}_c(\mathscr{E})$: THE ORBIT MUST REACH R

at this speed `b_max = 2.65 R` · Newton `2.21 R` · GR/Newton `1.198`
· at 220 km/s: `1010 R` , × `1.244` · slow limit 1/√(1−2C) = `1.244`

SUPPLY × PROBABILITY

What reaches the star, and *what stays*

$$ds^2 = -e^{\nu(r)}dt^2 + e^{\lambda(r)}dr^2 + r^2 d\Omega^2$$

THE TOV STAR: SLY, 1.40 M⊙, 11.70 km, C = 0.177

$$\frac{d^2\dot{N}_{\rm in}}{d\mathscr{E}\, d\mathscr{L}} = 8\pi^2\, m^3\, f(\mathscr{E})\, \mathscr{L}, \quad \mathscr{L} < \mathscr{L}_c(\mathscr{E})$$

PARTICLES ENTERING THE SURFACE, PER UNIT COORDINATE TIME

$$\dot{N}_{\rm X} = 8\pi^2 m^3 \iint P_{\rm cap}(\mathscr{E}, \mathscr{L})\, f(\mathscr{E})\, \mathscr{L}\, d\mathscr{L}\, d\mathscr{E}$$

THE CAPTURE RATE: THE KERNEL P_{cap} IS WHAT OBSCURA COMPUTES

$\mathscr{E}$ and $\mathscr{L}$ are the energy and angular momentum per unit mass, measured at infinity; between collisions they are **conserved**, and the particle follows the interior geodesic.

CAPTURED

$$\mathscr{E}_{\rm final} = \gamma_{\rm final}\, e^{\nu(r_{\rm coll})/2} < 1$$

after a collision the particle can no longer reach infinity: γ is its Lorentz factor for a static observer at the collision, $e^{\nu/2}$ the lapse there.

THE OPTICAL-DEPTH CLOSURE

A Poisson count and *three frozen choices*

$$P_{\mathrm{cap}} = 1 - e^{-\tau} \sum_{k < N_{\mathrm{min}}} \frac{\tau^k}{k!}$$

THE NUMBER OF COLLISIONS IS POISSON WITH MEAN T

$$\tau = 2 \int_{r_{\mathrm{tp}}}^{R} \frac{e^{(\nu+\lambda)/2}\, \Gamma_{n\chi}}{\sqrt{(\mathcal{E}^2 - V_{\mathrm{eff}}^2)}}\, dr$$

COLLISIONS ALONG THE INCIDENT GEODESIC, ONE CROSSING

The closure of Sirvente & Rueda (submitted): relativistic and covariant. OBSCURA was built to test it, not to replace it — and reduces to it exactly.

1 · FROZEN TRAJECTORY

Collisions change the energy, never the path: every particle keeps the incident geodesic.

2 · DETERMINISTIC LOSS

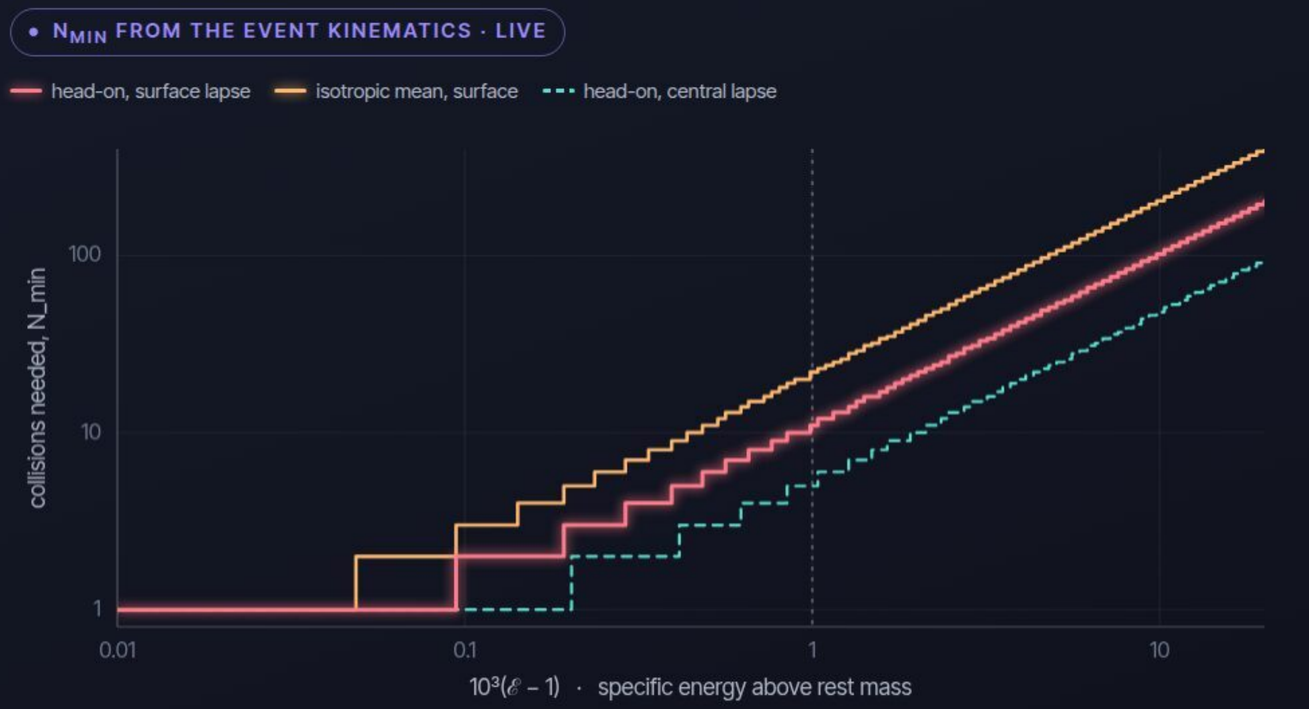
Each collision removes a fixed amount — head-on or the isotropic mean — so capture needs exactly N_{min} of them.

3 · EVALUATED AT THE SURFACE

N_{min} is computed with the lapse at R, wherever the collision actually happens.

100 KEV ON A NEUTRON OF 939 MEV

Capture decided in *a narrow window*



$$\frac{T}{K} = \frac{m_n m_\chi (\gamma + 1)(1 - \cos\theta^*)}{s}$$

KINETIC ENERGY GIVEN TO THE NEUTRON, EXACT

For $m_\chi \ll m_n$ each collision removes a fraction $\sim 2(m_\chi/m_n)(\gamma + 1) \approx 5\times10^{-4}$ at most, while it must lose up to a per cent of it: tens of collisions.

Deeper in the star the lapse is smaller and the same collision removes more of $\mathscr{E}$: N_{min} **halves** at the centre.

Slide the mass to the GeV range: one collision suffices — the WIMP limit.

THE TRANSPORT

One history, *collision by collision*

1. Draw a target optical depth $-\ln U$ and fly along the **interior geodesic** with $(\mathscr{E}, \mathscr{L})$ conserved, accumulating $d\tau = n_b \sigma\, dl$ (step R/4096).
2. At the collision, an **elastic two-body event** on a neutron at rest, isotropic in the centre of mass: energy loss and deflection from the same angles.
3. New γ and direction → new $(\mathscr{E}, \mathscr{L})$: the **orbit is rebuilt**.
4. **Captured** if $\mathscr{E} < 1$; **escaped** if it leaves unbound; otherwise fly again.

$$T = \frac{p^{*2}}{m_n}(1 - \cos\theta^*),\ p^* = \frac{m_n p_\chi}{\sqrt{s}}$$

EXACT, FROM THE INVARIANT t

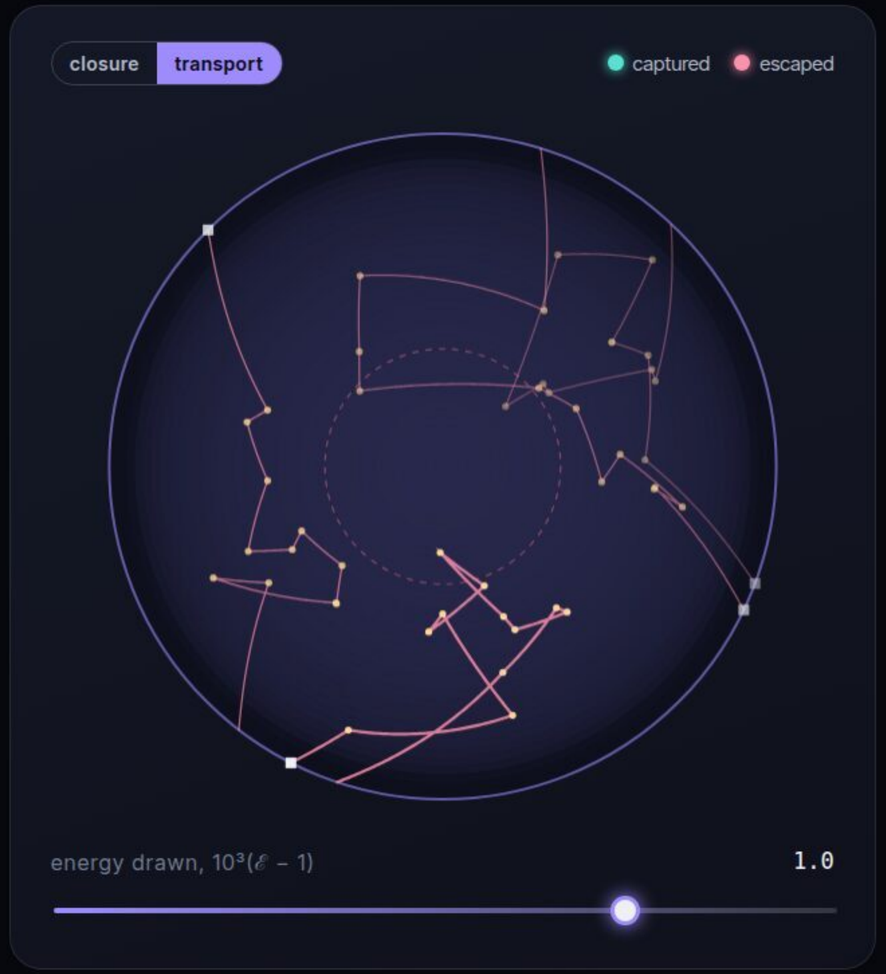
EVERY APPROXIMATION HAS A NAME

`GEO-8` interior tetrad transport · `COL-3` relativistic event · `PAULI-1` blocked rate · `STAT-1` Wilson intervals — each documented, cited in the source, and switchable.

Switch off the release, the sampling and the local lapse, and the Monte Carlo **is** the closure.

THREE MODES, ONE STAR · LIVE

Releasing the trajectory *broadens* the transition





REDUCTION

Switch it off, and it *is* the closure

SWITCH OFF	RECOVERS
trajectory update at collisions	the incident geodesic of τ
sampled energy loss	the recurrence behind N_{min}
loss at the collision radius	the cascade at the surface lapse
histories beyond one crossing	the single-crossing convention
neutrons at rest in the rate	not switched: up to 0.035 in P_{cap}

N_{min}

every capture of the realised closure happens at exactly N_{min}

$0.2_{\%}$

mean count vs $\sigma\int n_b dl$ by an independent quadrature

2.8σ

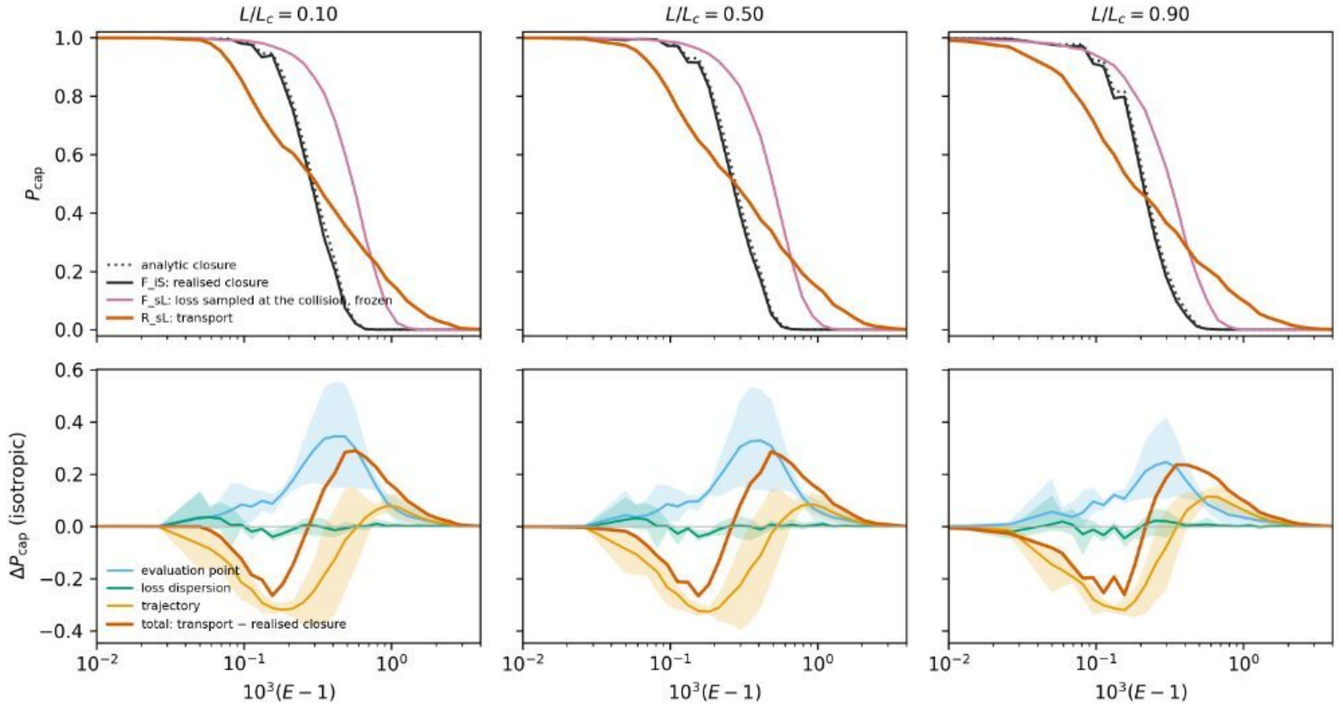
largest deviation of P_{cap} from the Poisson tail, 123 points × 5000 histories

10^{-10}

a sampled event vs an independent closed form; head-on = one step of the recurrence

THE DECOMPOSITION · $2^3 \times 2$ CONTROLS, ORDER-AVERAGED

Two effects of *opposite sign*



SWITCH	MEAN ΔP_{CAP}
• evaluation point	+0.09 … +0.14
• loss dispersion	$\lvert\cdot\rvert$ < 0.01
• trajectory	−0.09 … −0.11
total (signed mean)	+0.00 … +0.03
total, RMS over the transition	0.16 … 0.17

They **nearly cancel in the mean** but not in shape: the first shifts the transition up, the second broadens it. Single energies differ by up to **±0.29**.

An attribution with interactions, not three independent corrections.

Benchmark star, 100 keV, $\sigma = 10^{-44}$ cm^2, 5000 histories per point, no blocking.

Half the particles *turn back*

48–53%

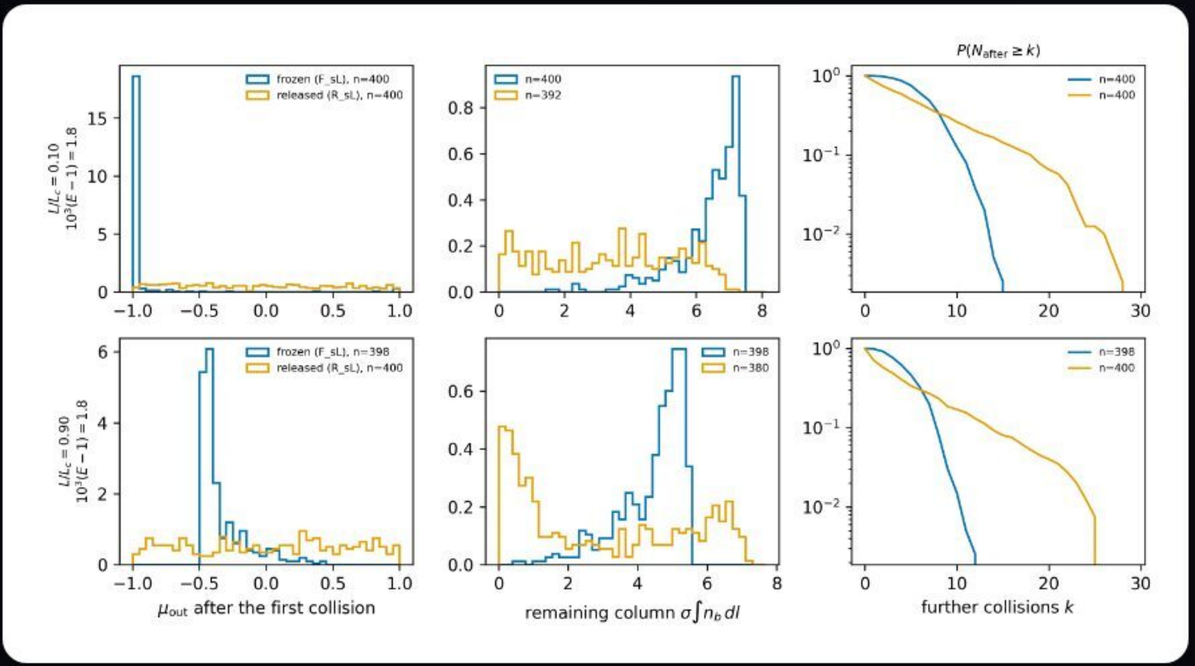
leave the first collision moving **outward** — frozen chord: 2–8%

≈ 3.1

mean column left to the surface, spread over 0.2–6.6 (10–90%) — frozen: 4.4–6.5, narrow

16–28%

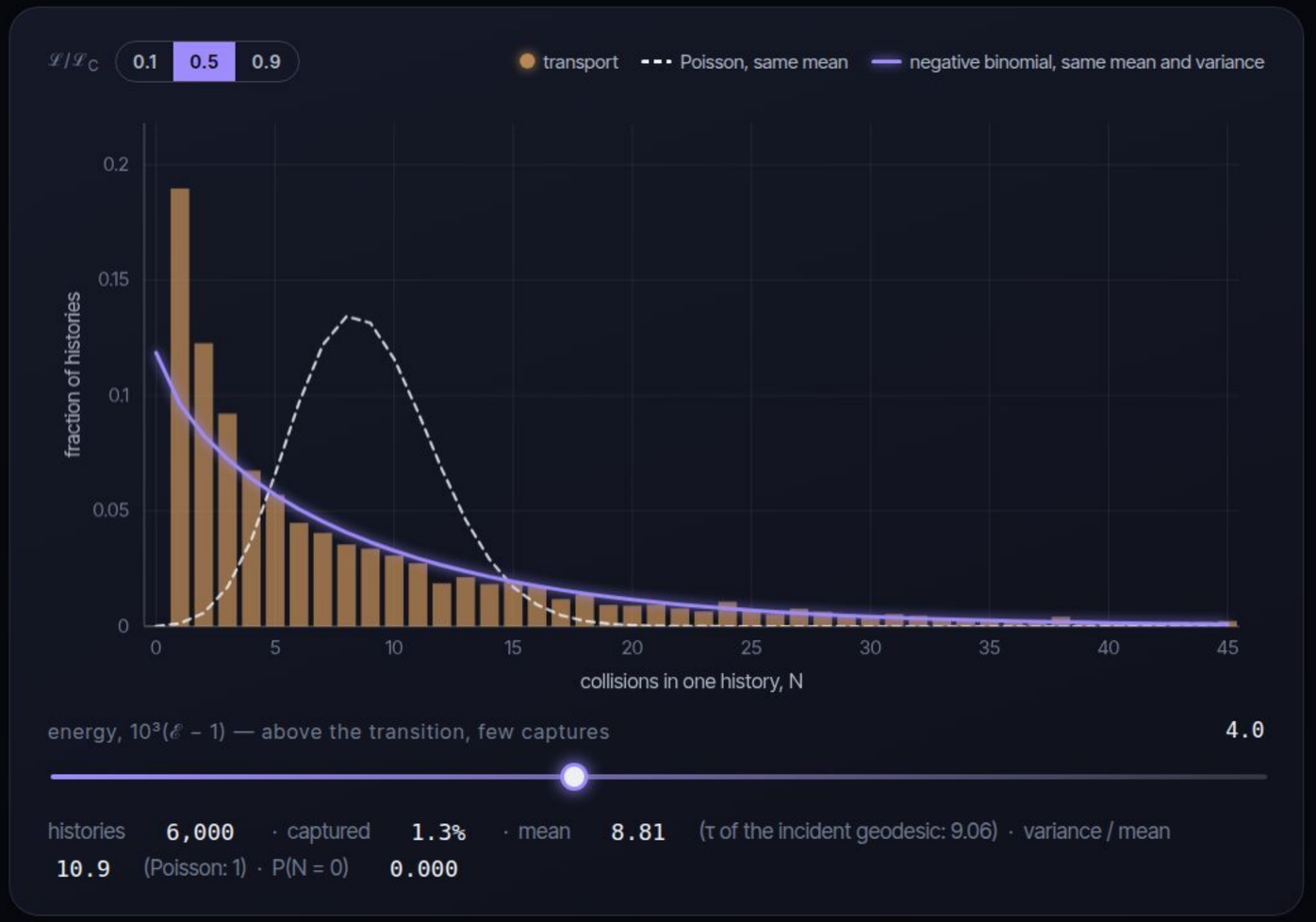
never collide again — frozen: at most 3%



Frozen (F_sL) vs released (R_sL) after the first collision, at $10^3(\mathscr{E} - 1) \approx 1.8$, for $\mathscr{L}/\mathscr{L}_c = 0.1$ and 0.9.

HOW MANY COLLISIONS · LIVE

A count that is *not Poisson*



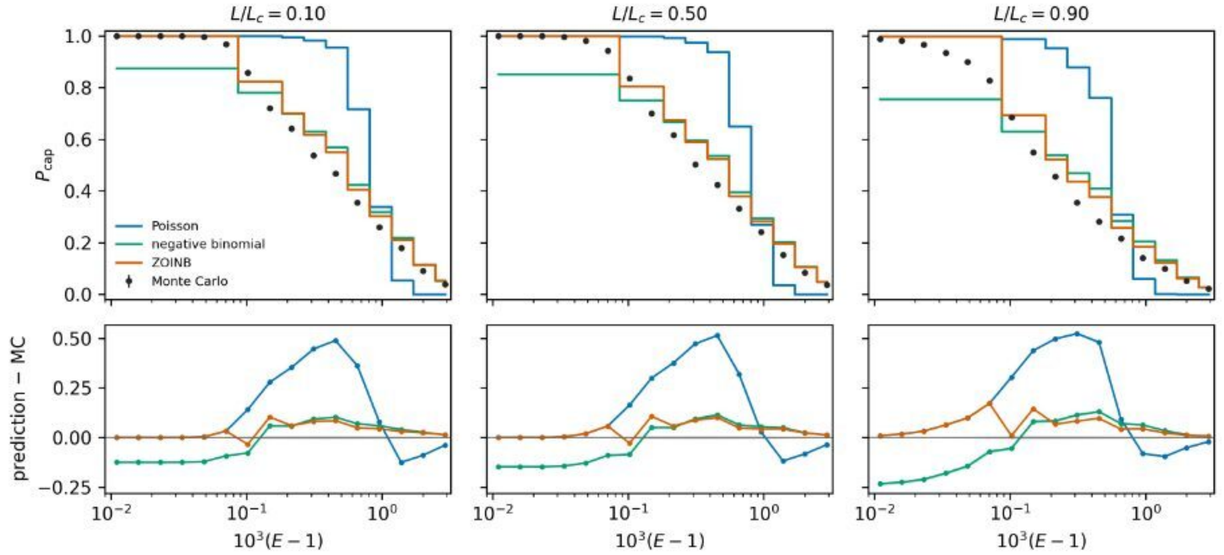
Above the transition almost nothing is captured, so each count is complete: the history ends when the particle leaves. The closure assumes Poisson with the mean τ. The **mean** is right; the **shape** is not.

$$P_{\mathrm{NB}}(k; r, p),\ r = \frac{m^2}{v - m}, p = \frac{m}{v}$$

POISSON COUNT WITH A GAMMA-DISTRIBUTED T

In the paper: variance **8–13 times** the mean. This is a random walk out of an optically thick sphere, not a straight chord.

Better laws, *honest residuals*



MEAN |PREDICTION − MC|	OUR SLY	BENCHMARK
Poisson, measured mean	0.15–0.19	0.16–0.18
negative binomial	0.08–0.11	0.057–0.064
zero-and-one-inflated NB	0.035–0.057	0.044–0.064

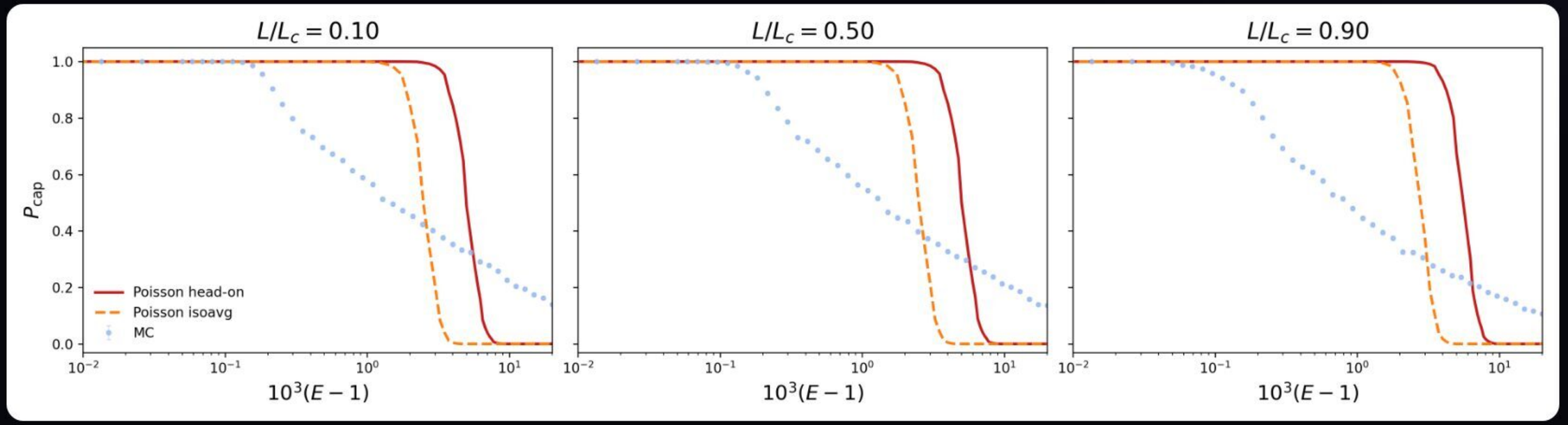
Calibration and test share **no histories**. The zero-and-one-inflated form removes the spurious zero-collision mass of the plain NB.

Residuals stay 6–23 times the combined Monte Carlo and calibration errors: closed forms that help, tested at fixed parameters — not yet a transferable law.

SLy 1.40 M☉, 11.70 km; calibrated above the transition on independent runs, tested inside it.

A MORE COMPACT STAR · SLY AT MAXIMUM MASS, 2.05 $M_\odot$

Broadening is *not a detail*



Points: the transport. Red: closure with head-on N_{min}; orange dashed: isotropic-mean N_{min}. 100 keV, $\sigma = 10^{-44}$ cm², no blocking.

Optical depth of the closure: $\tau \approx$ **25–27**, three times the canonical star.

The transport already falls below 0.95 at $10^3(\mathscr{E} - 1) = 0.1$–$0.2$, and keeps **11–14%** at 20, where both closures vanish.

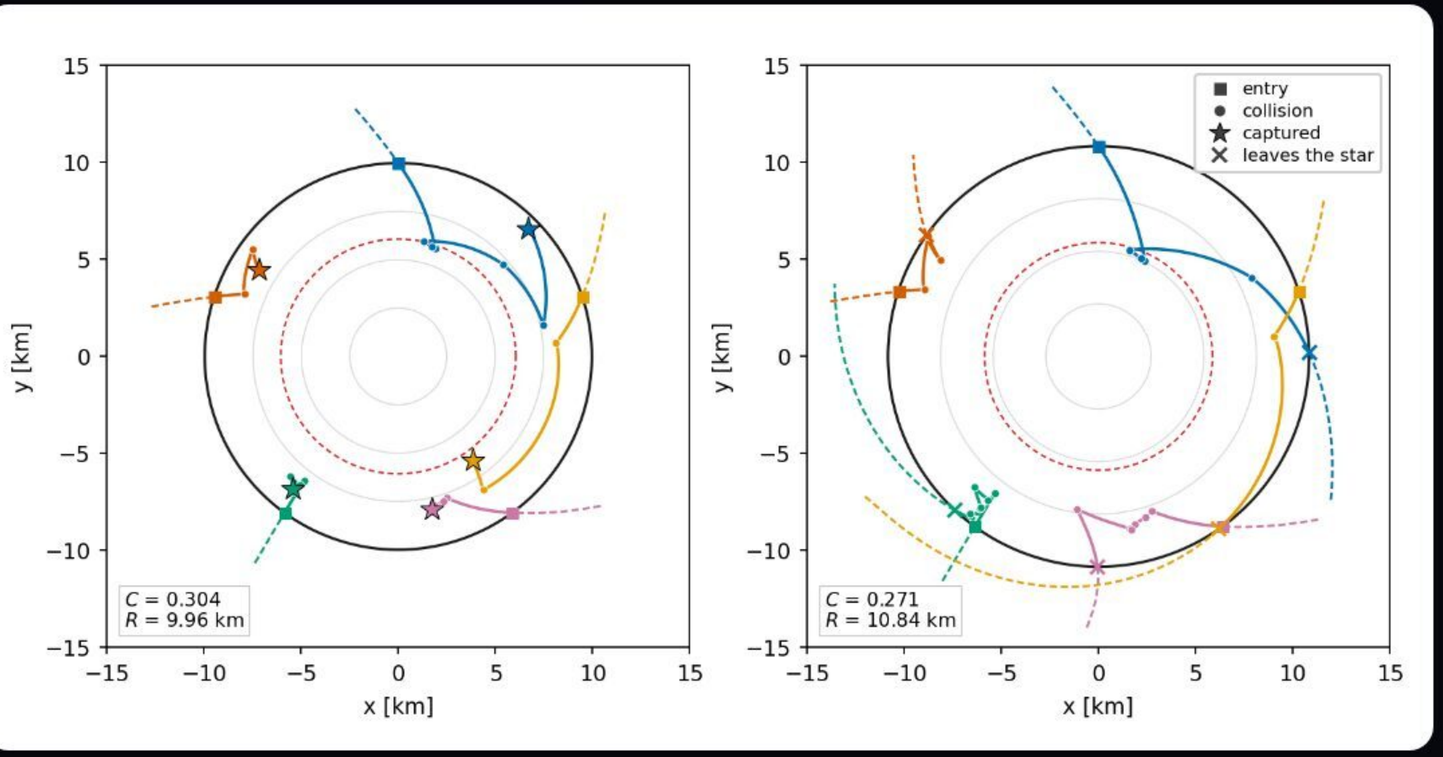
Mean |difference| from the isotropic-mean closure: **0.28–0.29**; single energies up to **0.6–0.7**.

SAME FIVE PARTICLES · DIFFERENT STARS

The equation of state *decides the fate*

SLY · 2.05 M⊙, 9.96 KM · 5/5 CAPTURED

SQM3 · 1.98 M⊙, 10.84 KM · 0/5 CAPTURED



Identical $\mathscr{E}$, $\mathscr{L}$, speed and impact parameter in both panels; m_χ = 100 keV, $\sigma = 10^{-44}$ cm². Re-traced geodesics, closing on every recorded collision: radius exact, angle from the entry point.

Five incident particles, the **same** phase-space points, sent into the maximum-compactness stars of a hadronic (SLy) and a strange-quark (SQM3) equation of state.

The same orbit crosses a different column and collides at different depths: $\mathscr{E}$ falls below 1 in SLy and not in SQM3, **five captures against none**.

Selected cases, not an average: over 50 000 histories, P_{cap} = **0.56** in SLy and **0.44** in SQM3.

BEYOND THE BENCHMARK

Pauli blocking and *cooling*

PAULI BLOCKING · IMPLEMENTED, NOT YET IN THE CAMPAIGNS

The recoiling neutron must leave the Fermi sea: the blocked optical depth is **7×10^{-5}** of the unblocked one.

But the collisions that do happen are violent: on average **57%** of the kinetic energy, against 2×10^{-4} for a static target.

So a blocked τ can never be paired with a static N_{min}: both must come from the **same blocked kernel**. The hierarchy has to be measured again, not rescaled.

Rate tabulated in ($\ln n_b$, $\ln(\gamma - 1)$) to 10^{-3}, the Fermi-surface shell of width $\sim3\times10^{-4}$ resolved by bisection.

COOLING TO 3KT/2 · AFTER CAPTURE

	10^7 K	10^8 K
collisions to capture	5	5
collisions to the threshold	4494	1374
re-entries into the star	396	191
coordinate time	5.1 s	6.8 s
time spent outside the star	97%	99%

Medians. A first-passage time in a cooling-only model — no heating, no detailed balance: **not** thermal equilibrium.

UNDER THE HOOD

Validated *stage by stage*

CHECK	AGREEMENT
interior transport vs independent orbit quadrature (path, azimuth)	$1–2\times10^{-4}$
P_{cap} with the step refined from $R/4096$ to $R/16384$	$\le 2\times10^{-4}$
exterior periapsis vs analytic turning point	$< 1.4\times10^{-13}$
GR / Newtonian focusing at $v \to 0$	1.244
surface hits: quadrant asymmetry, 10^6 samples	3×10^{-3}
CPU vs CUDA, 60 cases × 10^5 histories	4.5×10^{-3}
population equation vs $N = C_\chi t$	10^{-16}
Fermi EOS: Γ_{eff} at the two ends	$1.6667 \cdot 1.3333$



3

independent implementations of the N_{min} recurrence, same integers

425

assertions in 26 test suites

C++17 with OpenMP; the same engine compiled to WebAssembly for an interactive lab.

What the closure *misses*

- The Monte Carlo reduces **exactly** to the Poisson closure when its released ingredients are switched off.
- Where the loss is evaluated and whether the trajectory is released move P_{cap} by **±0.1** each, in opposite directions: right on average, **0.16–0.18 RMS** off in shape.
- Collision counts are strongly **overdispersed**; negative-binomial laws calibrated out of sample do much better than Poisson.
- Pauli blocking changes rate **and** kernel together: it must be measured, not rescaled.

> *A correct mean number of collisions is not enough for a Poisson closure.*
>
> the one-line summary of Part II

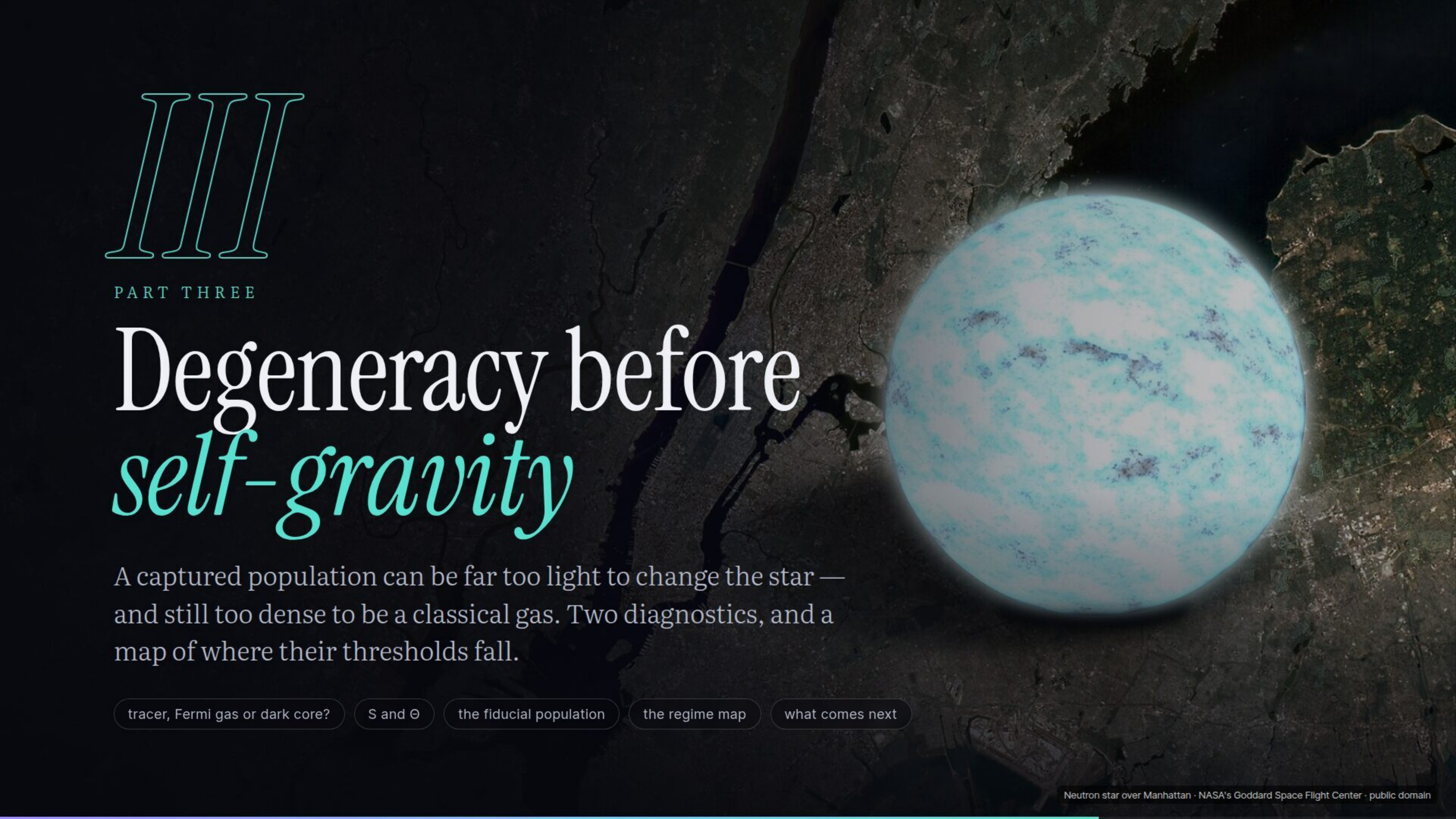
III

PART THREE

Degeneracy before *self-gravity*

A captured population can be far too light to change the star — and still too dense to be a classical gas. Two diagnostics, and a map of where their thresholds fall.

tracer, Fermi gas or dark core? | S and Θ | the fiducial population | the regime map | what comes next



Neutron star over Manhattan · NASA's Goddard Space Flight Center · public domain

THE QUESTION

After capture, *what is it?*

capture supplies $N_\chi(t)$ → **which regime?** → the right next model

A CLASSICAL TRACER

The usual assumption: the particles thermalise into a Boltzmann cloud in the fixed potential of the star. Right for a dilute gas.

$$n \propto e^{-m\Delta\Phi/kT}$$

A DEGENERATE TRACER

Fermions in a deep well can reach a phase-space density where **Pauli** matters — while still too light to change the gravity.

$$E_F \gtrsim kT$$

A DARK COMPONENT

Dark-matter admixed neutron stars: the dark fluid takes part in hydrostatic balance and shifts the mass–radius relation.

two-fluid TOV

TWO INDEPENDENT DIAGNOSTICS

One for gravity, *one for statistics*

$$S = \frac{M_\chi(r_\chi)}{M_b(r_\chi)}$$

SELF-GRAVITY · INSIDE THE REGION THE DARK MATTER OCCUPIES

S	REGIME	OMITTED GRAVITY S/(1 + S)
$< 10^{-3}$	tracer	$< 10^{-3}$
$10^{-3} – 10^{-1}$	transitional	up to 9%
$\geq 10^{-1}$	self-gravitating	≥ 9%

Local, not the global $M_\chi/M_\star$: $r_\chi = r_{90}$ holds 90% of the particles.

$$\Theta = \frac{T}{T_F} = \frac{k_B T}{E_F}$$

CLASSICALITY · THERMAL ENERGY OVER FERMI ENERGY

Θ	REGIME	FERMI SHARE 1/(1 + Θ)
> 10	thermal	< 0.09
0.1 – 10	transitional	0.09 – 0.91
< 0.1	degenerate	> 0.91

There is no general reason for the two sets of thresholds to **coincide**.

THE TWO CLOSURES

A Boltzmann cloud, and *the Fermi limit*

$$n_\chi(r) = n_0 \exp\left[-\frac{m_\chi \Delta\Phi(r)}{k_B T}\right]$$

THERMAL TRACER IN THE FIXED POTENTIAL OF THE STAR

$$N_\chi = 4\pi \int_0^R n_\chi r^2 \sqrt{g_{rr}}\, dr$$

FIXES n_0 · r_{90} HOLDS 0.9 N_χ

$$p_F = \hbar\,(3\pi^2 n_\chi)^{1/3},\ x_F = \frac{p_F}{m_\chi c}$$

SPIN ½, g = 2

$$E_F = m_\chi c^2\,(\sqrt{(1 + x_F^2)} - 1)$$

$P \propto \rho^{5/3}$ for $x_F \ll 1$ · $P \propto \rho^{4/3}$ for $x_F \gg 1$

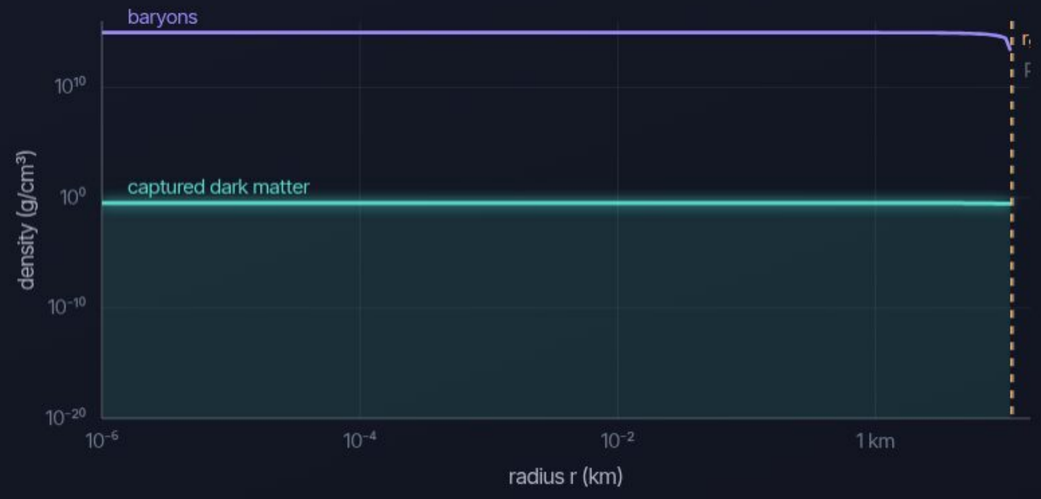
The cloud has no edge: r_{90} ignores the dilute tail and measures where the particles actually are.

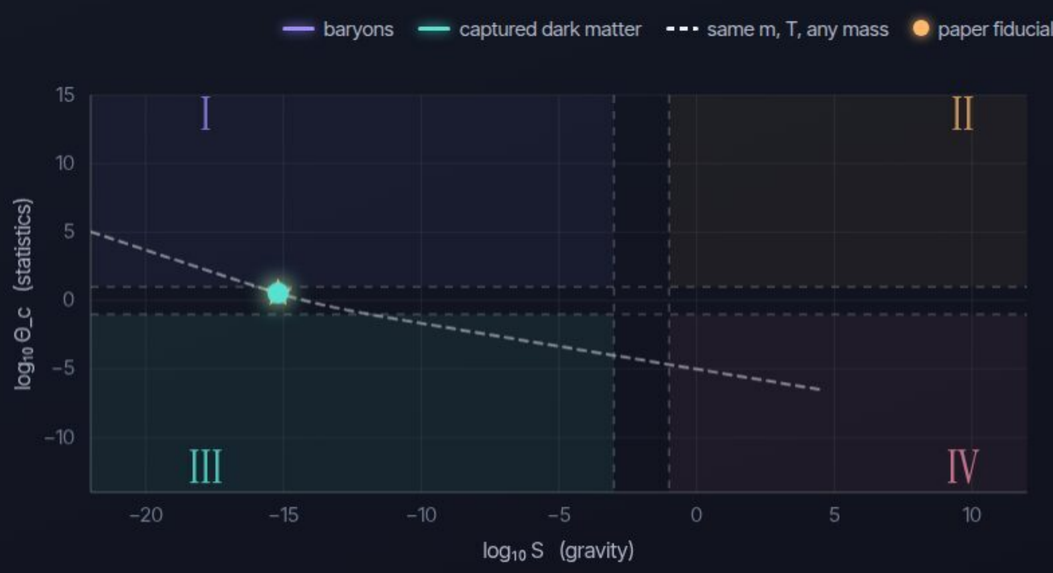
Θ is a **diagnostic ratio**, not a finite-temperature chemical potential: it says when a Fermi–Dirac treatment is needed.

THE TWO DIAGNOSTICS · LIVE

Move a cloud through *the plane*

PRESETS | **paper fiducial · 100 keV, 10^9 K** | 1 GeV WIMP, 10^5 K | 1 keV, 10^6 K | 10^6 GeV, 10^5 K





Dashed lines: the thresholds $S = 10^{-3}, 10^{-1}$ and $\Theta_c = 10, 0.1$. The bands between them are transitional.

I Boltzmann tracer $S < 10^{-3}, \Theta_c > 10$

II Boltzmann, self-gravitating $S \geq 10^{-1}, \Theta_c > 10$

III Fermi tracer $S < 10^{-3}, \Theta_c < 0.1$

IV Fermi, self-gravitating $S \geq 10^{-1}, \Theta_c < 0.1$

particle mass m_χ **100 keV** | core temperature **1.0×10^9 K** | captured mass M_χ **2.36×10^{18} g**

r_{90} **11.3 km** · central density **0.328 g/cm³** · x_F **0.748** · E_F **24.9 keV** vs kT **86.2 keV** · S **6.37×10^{-16}** **TRACER** · Θ_c **3.46** **TRANSITIONAL**

· on this line $\Theta_c = 1$ at **$S \approx 10^{-14}$**

Pauli comes *first* for light particles

A compact cloud sits in the harmonic core, $\Delta\Phi = (2\pi/3)G\rho_c r^2$. Its size cancels between S and the central density:

$$n_0 \approx 4.6\, S\, \frac{\rho_c}{m_\chi}$$

INDEPENDENT OF T

$$\Theta_c \propto T\, m_\chi^{5/2}\, S^{-2/3} \Rightarrow S(\Theta_c = 1) \propto \frac{m_\chi^{5/2}\, T^{3/2}}{\rho_c}$$

NON-RELATIVISTIC FERMI GAS

Light particles are many per gram: they fill phase space long before they weigh anything.

PARTICLE, CORE	S AT $\Theta_c = 1$
1 keV, 10^6 K	10^{-24}
100 keV, 10^9 K	10^{-14}
1 GeV, 10^5 K	10^{-11}
1 GeV, 10^7 K	10^{-8}
1 TeV, 10^5 K	10^{-4}

Self-gravity starts at $\mathbf{S \sim 10^{-3}}$. For anything lighter than about a TeV, the gas turns **degenerate first**.

Full thermal profile in the SLy star, as in the previous panel.

THE PIPELINE

From the star to *the diagnostics*

baryonic TOV star → $b_{max}(V)$ → Γ_{inc} → f_{cap} (Monte Carlo) → C_χ → $N_\chi(t)$ → $\rho_\chi(r)$ → (S, Θ)

$$\Gamma_{inc} = n_\chi \pi \int b_{max}^2 v\, f(v)\, dv$$

FOCUSED FLUX ONTO THE SURFACE

$$C_\chi = \Gamma_{inc}\, f_{cap}$$

f_{cap}: CAPTURED OVER SAMPLED

$$\frac{dN_\chi}{dt} = C_\chi - C_A N_\chi^2 - C_E N_\chi$$

ANNIHILATION, EVAPORATION: OFF IN THE FIDUCIAL

The order matters: capture supplies the particle budget, S tests whether the fixed gravitational background is adequate, Θ tests whether the classical closure is.

Gravitationally invisible, *not quite classical*

	MAXWELL–BOLTZMANN	RAR–FD
incidence rate Γ_{inc}	4.36×10^{29} s^{-1}	2.33×10^{29} s^{-1}
capture fraction f_{cap}	0.962	0.959
captured particles N_χ	1.32×10^{46}	7.06×10^{45}
captured mass M_χ	2.36×10^{18} g	1.26×10^{18} g
cloud radius	11.3 km (r_{90})	11.7 km (support)
self-gravity S	6.4×10^{-16}	3.8×10^{-16}
classicality Θ_c	3.46	5.07

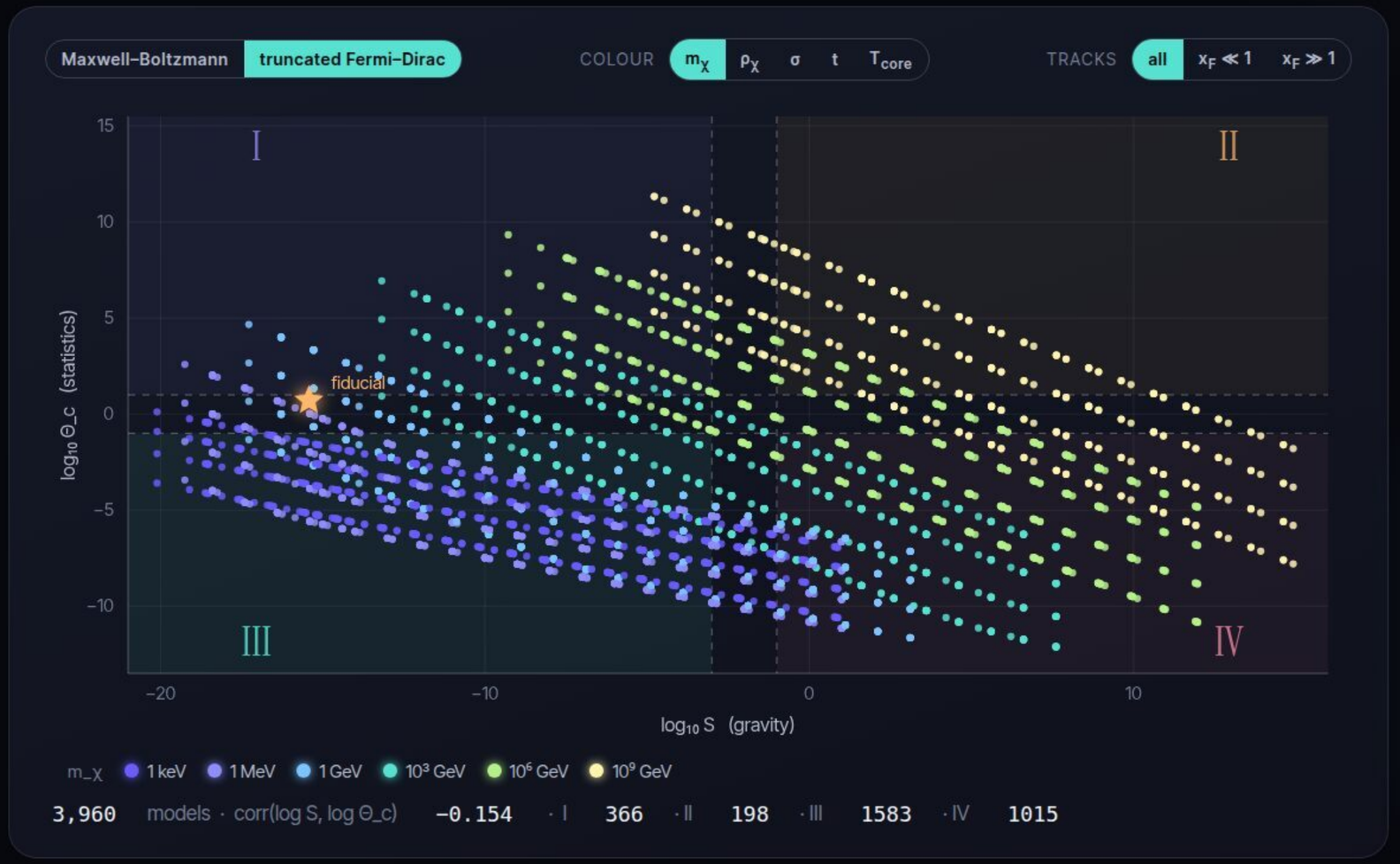
Fiducial: $m_\chi = 100$ keV, $\sigma = 10^{-44}$ cm^2, $\rho_\chi = 0.4$ GeV cm^{-3}, $T_{core} = 10^9$ K, $t = 10^9$ yr, SLy 1.40 $M_\odot$.

$M_\chi/M_\star \approx 10^{-15}$: no mass–radius shift to look for. Yet $\Theta_c \approx 3$–5 is in the **transitional** band: corrections of order one to Boltzmann.

Changing the halo from MB to RAR–FD changes the **supply** by 47% and the microscopic capture by 0.3%.

THE REGIME MAP · EVERY MODEL OF THE SURVEY

The two boundaries *do not coincide*



I

BOLTZMANN TRACER

the usual assumption

II

BOLTZMANN, SELF-GRAVITATING

heavy, classical, massive

III

FERMI TRACER

degenerate, gravitationally passive

IV

FERMI, SELF-GRAVITATING

toward a dark core

READING THE MAP

Two slopes, *one Fermi energy*

Hold m_χ, σ, t, T fixed and change ρ_χ: the profile keeps its shape, $S \propto n_0 \propto N_\chi$, and $p_F \propto n_0^{1/3}$:

$$\Theta_c \propto S^{-2/3}$$

$x_F \ll 1 \cdot E_F \approx p_F^2/2m$

$$\Theta_c \propto S^{-1/3}$$

$x_F \gg 1 \cdot E_F \approx p_F c$

RAR–FD SURVEY	TRACKS	POINTS	MEDIAN SLOPE
non-relativistic family	352	2112	−0.667
relativistic family	213	1278	−0.334

THE MASS TREND

Heavier particles are less degenerate: $d \log \Theta_c / d \log m_\chi \approx$ **0.49** (MB) and **0.42** (RAR–FD). The degenerate quadrants hold **85–89%** of the lightest models and **21–28%** of the heaviest.

NOT A CHANGE OF PARTICLE

III → IV is the same Fermi gas becoming heavier where it sits: the growth of S, not of the particle mass.

MB vs RAR–FD: corr($\log S$, $\log \Theta_c$) = −0.652 vs −0.154.

WHAT COMES NEXT

The right model *for each quadrant*

I

$S \ll 1,\ \Theta \gg 1$

a Boltzmann cloud in the fixed star — the standard assumption, justified

II

$S \gtrsim 0.1,\ \Theta \gg 1$

classical, but it shapes its own potential

III

$S \ll 1,\ \Theta \lesssim 1$

a capture-normalised **Fermi tracer** in the fixed star — finite-T if $\Theta \sim 1$–10

IV

$S \gtrsim 0.1,\ \Theta \ll 1$

only here: a **two-fluid TOV**, the dark-matter admixed star

Capture calculations and admixed-star models are not interchangeable endpoints. A small $M_\chi/M_\star$ or S says the structure is unchanged; it does **not** justify a Boltzmann profile.

NOT YET INCLUDED

Annihilation heating, collapse and black-hole formation, rotation, magnetic fields, and a self-consistent time evolution of the captured Fermi gas. Θ is a diagnostic, not a finite-temperature solution.

CONCLUSIONS

Capture, *then character*

- **OBSCURA** follows dark-matter collision histories through the interior metric and reduces exactly to the optical-depth closure.
- At 100 keV, the frozen trajectory and the surface evaluation each move P_{cap} by **~0.1**, in opposite directions; the count of collisions is **overdispersed**.
- After capture, **S** and **Θ** answer different questions: the fiducial population is an extreme tracer ($S \sim 10^{-16}$) in the transitional band ($\Theta_c \approx 3$–5).
- For light particles **degeneracy comes before self-gravity**: the thermalised-WIMP assumption has to be tested in (S, Θ), not by $M_\chi/M_\star$.

NEXT STEPS

- Pauli-blocked capture campaigns with the consistent blocked kernel.
- Capture rates over the full incident phase space, RAR halo included.
- Thermalisation with heating and detailed balance, then a finite-T Fermi tracer.
- Two-fluid structure once S leaves the tracer regime.

The panels in this seminar run in the browser, on the same physics as OBSCURA: *rafael-lima.pages.dev/talks*.

IMAGE CREDITS

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Paper figures — de Lima & Rueda, *A Monte Carlo relativistic framework for dark matter capture by neutron stars* (in preparation); de Lima & Rueda, *Degeneracy Before Self-Gravity: Dark Matter Regimes in Neutron Stars* (draft)

Data in Part I — RAR model: solved in the browser after Argüelles et al., PDU 21, 82 (2018); rotation curve: SPARC (Lelli, McGaugh & Schombert 2016); CMB: Planck 2018 and CLASS v3.4.1 (Blas, Lesgourgues & Tram 2011); direct-detection and axion limits: compilations by C. O'Hare (DirectDetectionPlots; AxionLimits, Zenodo 3932430), neutrino fog: O'Hare 2021

SLy × SQM3 histories — OBSCURA, re-traced along their geodesics

Regime-map panel — the survey data of the second paper, from the OBSCURA pipeline

Neutron star — SLy equation of state (Douchin & Haensel 2001) in the piecewise-polytropic fit of Read, Lackey, Owen & Friedman, PRD 79, 124032 (2009)

OBSCURA logo — OBSCURA project

THANK YOU

Questions?

Slides and live panels, on the same physics as OBSCURA, at
rafael-lima.pages.dev/talks

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